

Main

December 11, 2025

```
[143]: # Load data
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import numpy as np
df = pd.read_csv("../data/train.csv")
```

```
/var/folders/hs/fgtpkc7n7gg59wjygcjmqv80000gn/T/ipykernel_33457/4040572339.py:6
: DtypeWarning: Columns (26) have mixed types. Specify dtype option on import or
set low_memory=False.
df = pd.read_csv("../data/train.csv")
```

```
[144]: # Check the dataset
print(df.info())
print(df.describe())
# Check the data of each column
print("\nData sample:\n", df.head())
print("\nMissing values per column:\n", df.isnull().sum().
↪sort_values(ascending=False).head(10))
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 100000 entries, 0 to 99999
Data columns (total 28 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   ID                                     100000 non-null object
1   Customer_ID                           100000 non-null object
2   Month                                 100000 non-null object
3   Name                                  90015 non-null  object
4   Age                                   100000 non-null object
5   SSN                                   100000 non-null object
6   Occupation                            100000 non-null object
7   Annual_Income                         100000 non-null object
8   Monthly_Inhand_Salary                 84998 non-null  float64
9   Num_Bank_Accounts                     100000 non-null int64
10  Num_Credit_Card                        100000 non-null int64
11  Interest_Rate                          100000 non-null int64
12  Num_of_Loan                            100000 non-null object
13  Type_of_Loan                           88592 non-null  object
```

14	Delay_from_due_date	100000	non-null	int64
15	Num_of_Delayed_Payment	92998	non-null	object
16	Changed_Credit_Limit	100000	non-null	object
17	Num_Credit_Inquiries	98035	non-null	float64
18	Credit_Mix	100000	non-null	object
19	Outstanding_Debt	100000	non-null	object
20	Credit_Utilization_Ratio	100000	non-null	float64
21	Credit_History_Age	90970	non-null	object
22	Payment_of_Min_Amount	100000	non-null	object
23	Total_EMI_per_month	100000	non-null	float64
24	Amount_invested_monthly	95521	non-null	object
25	Payment_Behaviour	100000	non-null	object
26	Monthly_Balance	98800	non-null	object
27	Credit_Score	100000	non-null	object

dtypes: float64(4), int64(4), object(20)

memory usage: 21.4+ MB

None

	Monthly_Inhand_Salary	Num_Bank_Accounts	Num_Credit_Card	\
count	84998.000000	100000.000000	100000.000000	
mean	4194.170850	17.091280	22.47443	
std	3183.686167	117.404834	129.05741	
min	303.645417	-1.000000	0.000000	
25%	1625.568229	3.000000	4.000000	
50%	3093.745000	6.000000	5.000000	
75%	5957.448333	7.000000	7.000000	
max	15204.633333	1798.000000	1499.000000	

	Interest_Rate	Delay_from_due_date	Num_Credit_Inquiries	\
count	100000.000000	100000.000000	98035.000000	
mean	72.466040	21.068780	27.754251	
std	466.422621	14.860104	193.177339	
min	1.000000	-5.000000	0.000000	
25%	8.000000	10.000000	3.000000	
50%	13.000000	18.000000	6.000000	
75%	20.000000	28.000000	9.000000	
max	5797.000000	67.000000	2597.000000	

	Credit_Utilization_Ratio	Total_EMI_per_month
count	100000.000000	100000.000000
mean	32.285173	1403.118217
std	5.116875	8306.041270
min	20.000000	0.000000
25%	28.052567	30.306660
50%	32.305784	69.249473
75%	36.496663	161.224249
max	50.000000	82331.000000

Data sample:

	ID	Customer_ID	Month	Name	Age	SSN	Occupation	\
0	0x1602	CUS_0xd40	January	Aaron Maashoh	23	821-00-0265	Scientist	
1	0x1603	CUS_0xd40	February	Aaron Maashoh	23	821-00-0265	Scientist	
2	0x1604	CUS_0xd40	March	Aaron Maashoh	-500	821-00-0265	Scientist	
3	0x1605	CUS_0xd40	April	Aaron Maashoh	23	821-00-0265	Scientist	
4	0x1606	CUS_0xd40	May	Aaron Maashoh	23	821-00-0265	Scientist	

	Annual_Income	Monthly_Inhand_Salary	Num_Bank_Accounts	...	Credit_Mix	\
0	19114.12	1824.843333	3	...	-	
1	19114.12	NaN	3	...	Good	
2	19114.12	NaN	3	...	Good	
3	19114.12	NaN	3	...	Good	
4	19114.12	1824.843333	3	...	Good	

	Outstanding_Debt	Credit_Utilization_Ratio	Credit_History_Age	\
0	809.98	26.822620	22 Years and 1 Months	
1	809.98	31.944960	NaN	
2	809.98	28.609352	22 Years and 3 Months	
3	809.98	31.377862	22 Years and 4 Months	
4	809.98	24.797347	22 Years and 5 Months	

	Payment_of_Min_Amount	Total_EMI_per_month	Amount_invested_monthly	\
0	No	49.574949	80.41529543900253	
1	No	49.574949	118.28022162236736	
2	No	49.574949	81.699521264648	
3	No	49.574949	199.4580743910713	
4	No	49.574949	41.420153086217326	

	Payment_Behaviour	Monthly_Balance	Credit_Score
0	High_spent_Small_value_payments	312.49408867943663	Good
1	Low_spent_Large_value_payments	284.62916249607184	Good
2	Low_spent_Medium_value_payments	331.2098628537912	Good
3	Low_spent_Small_value_payments	223.45130972736786	Good
4	High_spent_Medium_value_payments	341.48923103222177	Good

[5 rows x 28 columns]

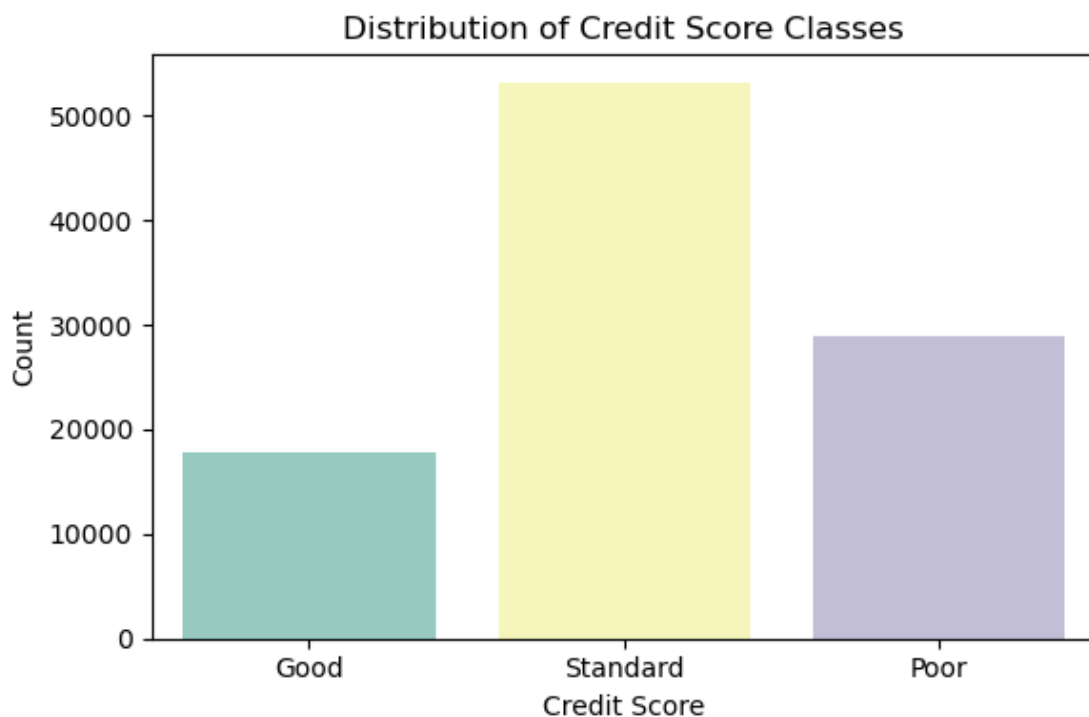
Missing values per column:

Monthly_Inhand_Salary	15002
Type_of_Loan	11408
Name	9985
Credit_History_Age	9030
Num_of_Delayed_Payment	7002
Amount_invested_monthly	4479
Num_Credit_Inquiries	1965
Monthly_Balance	1200
ID	0
Changed_Credit_Limit	0

dtype: int64

```
[145]: # EDA Visualize distributions of key features

# Distribution of target variable
#1 Target variable distribution
plt.figure(figsize=(6,4))
sns.countplot(data=df, x="Credit_Score", hue="Credit_Score", palette="Set3")
plt.title("Distribution of Credit Score Classes")
plt.xlabel("Credit Score")
plt.ylabel("Count")
plt.tight_layout()
plt.show()# Explore feature distributions by Credit Score
```



```
[146]: #Distributions of key features
# 1 Annual Income distribution
df_plot = df.copy()

df_plot["Annual_Income"] = (
    df_plot["Annual_Income"].astype(str)
    .str.replace(",", "", regex=False)
    .str.replace("_", "", regex=False)
)
```

```

df_plot["Annual_Income"] = pd.to_numeric(df_plot["Annual_Income"],
↳errors="coerce")

cap_value = df_plot["Annual_Income"].quantile(0.99)
df_plot["Annual_Income"] = np.minimum(df_plot["Annual_Income"], cap_value)

plt.figure(figsize=(6,4))
sns.histplot(
    data=df_plot,
    x="Annual_Income",
    kde=True,
    bins=30,
    color="skyblue"
)
plt.title("Distribution of Annual Income (Capped)")
plt.xlabel("Annual Income (USD)")
plt.ylabel("Count")
plt.tight_layout()
plt.show()

# 2 Credit Utilization Ratio distribution
df_plot = df.copy()
df_plot["Credit_Utilization_Ratio"] = pd.
↳to_numeric(df_plot["Credit_Utilization_Ratio"], errors="coerce")

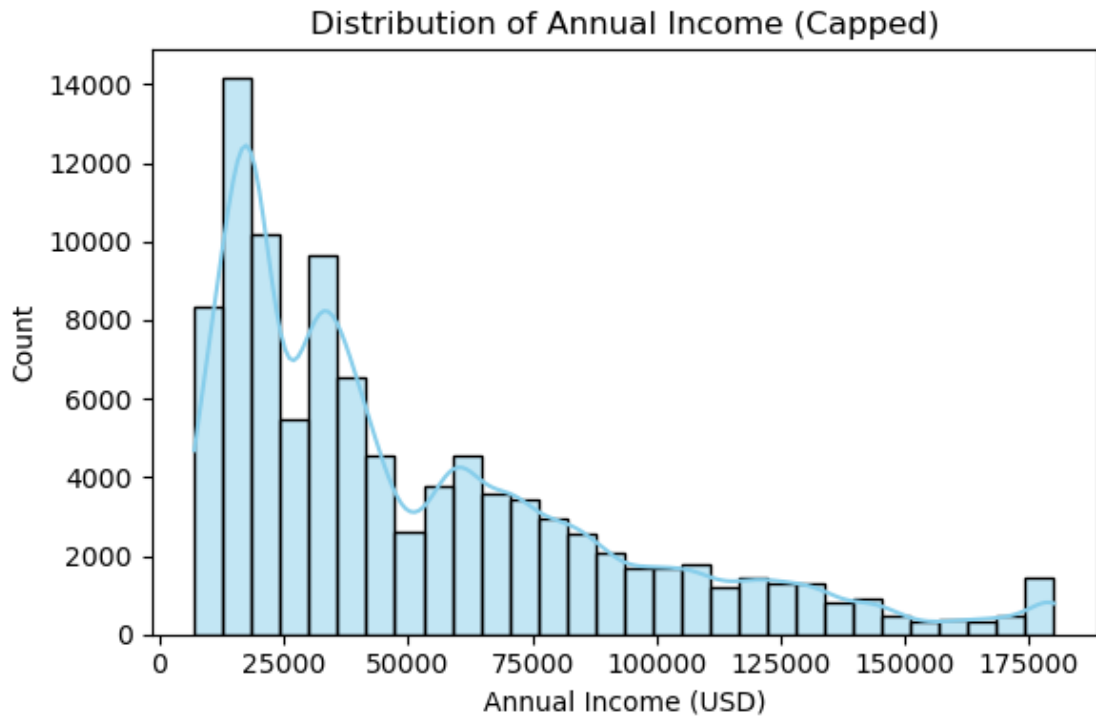
cap = df_plot["Credit_Utilization_Ratio"]
df_plot["Credit_Utilization_Ratio"] = np.
↳minimum(df_plot["Credit_Utilization_Ratio"], cap)

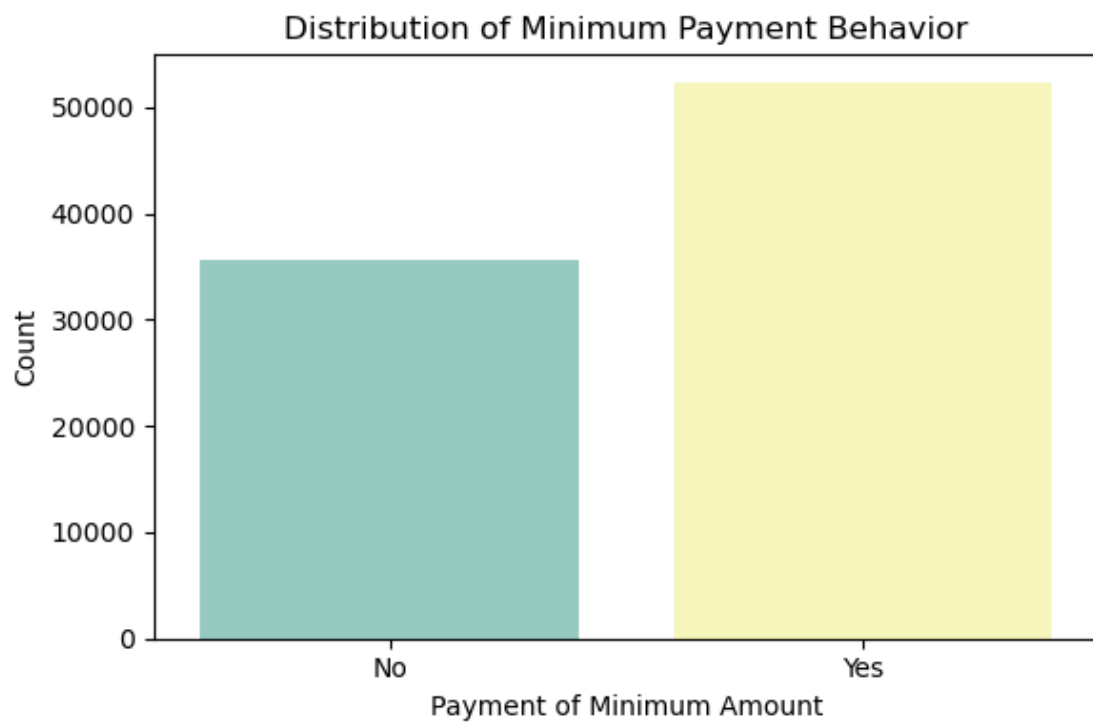
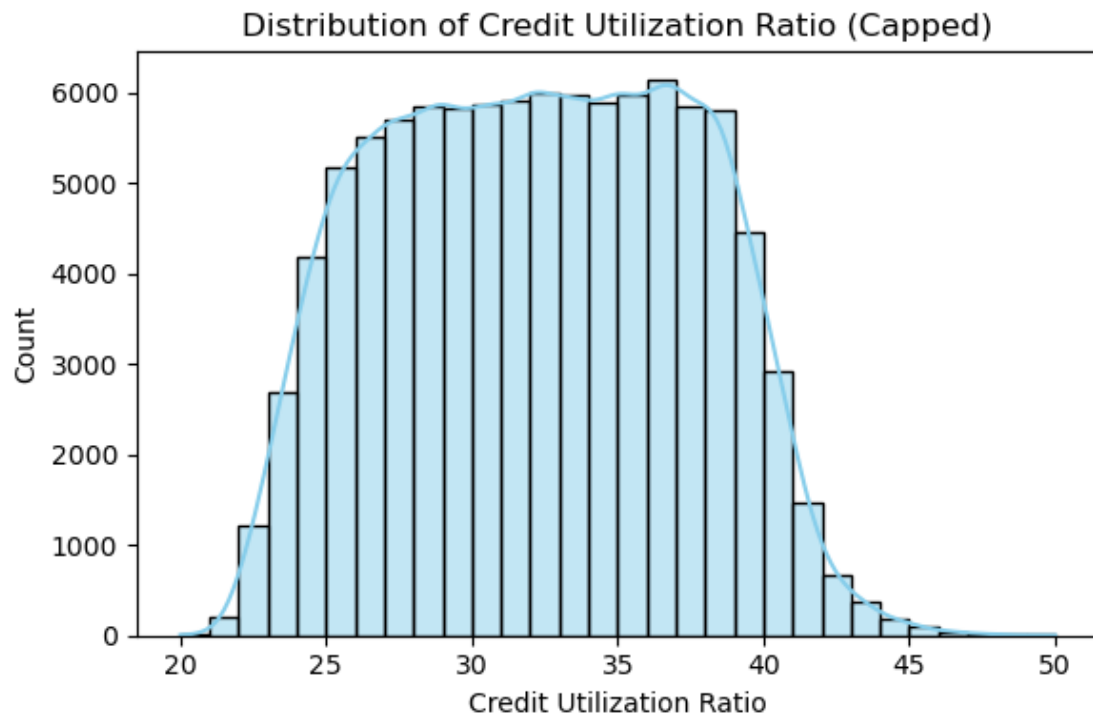
plt.figure(figsize=(6,4))
sns.histplot(
    data=df_plot,
    x="Credit_Utilization_Ratio",
    kde=True,
    bins=30,
    color="skyblue"
)
plt.title("Distribution of Credit Utilization Ratio (Capped)")
plt.xlabel("Credit Utilization Ratio")
plt.ylabel("Count")
plt.tight_layout()
plt.show()

# 3 Payment of Minimum Amount distribution
df_eda = df[df["Payment_of_Min_Amount"].isin(["Yes", "No"])]

```

```
plt.figure(figsize=(6,4))
sns.countplot(data=df_eda, x="Payment_of_Min_Amount",
             hue="Payment_of_Min_Amount", palette="Set3")
plt.title("Distribution of Minimum Payment Behavior")
plt.xlabel("Payment of Minimum Amount")
plt.ylabel("Count")
plt.tight_layout()
plt.show()
```





```

[147]: # Feature distributions by Credit Score
# 1 Age distribution by Credit Score
df_plot = df.copy()

df_plot["Age"] = pd.to_numeric(df_plot["Age"], errors="coerce")

df_plot = df_plot[(df_plot["Age"] > 0) & (df_plot["Age"] < 100)]

plt.figure(figsize=(6,4))
sns.boxplot(
    data=df_plot,
    x="Credit_Score",
    hue="Credit_Score",
    y="Age",
    palette="Set3",
    order=["Good", "Standard", "Poor"]
)
plt.title("Age Distribution by Credit Score")
plt.xlabel("Credit Score")
plt.ylabel("Age")
plt.tight_layout()
plt.show()

# 2 Number of Bank Accounts distribution by Credit Score
df_plot = df.copy()
df_plot["Num_Bank_Accounts"] = pd.to_numeric(df_plot["Num_Bank_Accounts"],
    errors="coerce")
df_plot = df_plot[df_plot["Num_Bank_Accounts"] >= 0]

plt.figure(figsize=(6,4))
sns.boxplot(
    data=df_plot,
    x="Credit_Score",
    hue="Credit_Score",
    y="Num_Bank_Accounts",
    order=["Good", "Standard", "Poor"],
    palette="Set3",
    width=0.5
)

plt.title("Number of Bank Accounts by Credit Score (Zoomed In)", fontsize=15,
    weight='bold')
plt.xlabel("Credit Score Category", fontsize=12)
plt.ylabel("Number of Bank Accounts", fontsize=12)

plt.ylim(-1, 20)

```



```

sns.despine()
plt.tight_layout()
plt.show()

# 3 Outstanding Debt distribution by Credit Score
df_plot = df.copy()

df_plot["Outstanding_Debt"] = (
    df_plot["Outstanding_Debt"]
    .astype(str)
    .str.replace(",", "", regex=False)
    .replace("NA", np.nan)
)

df_plot["Outstanding_Debt"] = pd.to_numeric(df_plot["Outstanding_Debt"],
    ↪errors="coerce")

plt.figure(figsize=(6,4))
sns.violinplot(
    data=df_plot,
    y="Credit_Score",
    x="Outstanding_Debt",
    hue="Credit_Score",
    order=["Good", "Standard", "Poor"],
    palette="Set3",
    inner="quartile"
)

plt.title("Outstanding Debt by Credit Score", fontsize=15, weight='bold')
plt.xlabel("Outstanding Debt (dollars)", fontsize=12)
plt.ylabel("Credit Score Category", fontsize=12)

sns.despine()
plt.tight_layout()
plt.show()

# 4 Number of Delayed Payment distribution by Credit Score
df_plot = df.copy()
df_plot["Num_of_Delayed_Payment"] = pd.
    ↪to_numeric(df_plot["Num_of_Delayed_Payment"], errors="coerce")

cap = df_plot["Num_of_Delayed_Payment"].quantile(0.98)
df_plot["Num_of_Delayed_Payment"] = np.
    ↪minimum(df_plot["Num_of_Delayed_Payment"], cap)

```

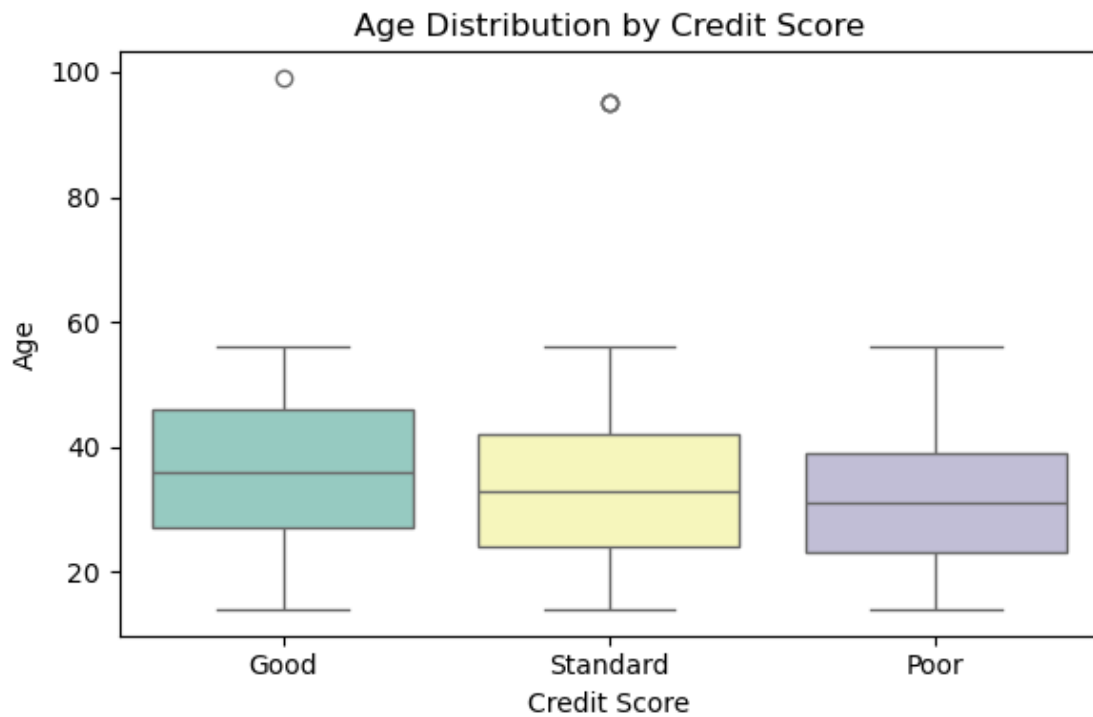
```

plt.figure(figsize=(6,4))
sns.boxplot(
    data=df_plot,
    x="Credit_Score",
    hue="Credit_Score",
    y="Num_of_Delayed_Payment",
    palette="Set3",
    order=["Good", "Standard", "Poor"]
)
plt.title("Delayed Payments by Credit Score (Capped at 98th Percentile)")
plt.xlabel("Credit Score")
plt.ylabel("Number of Delayed Payments")
plt.tight_layout()
plt.show()

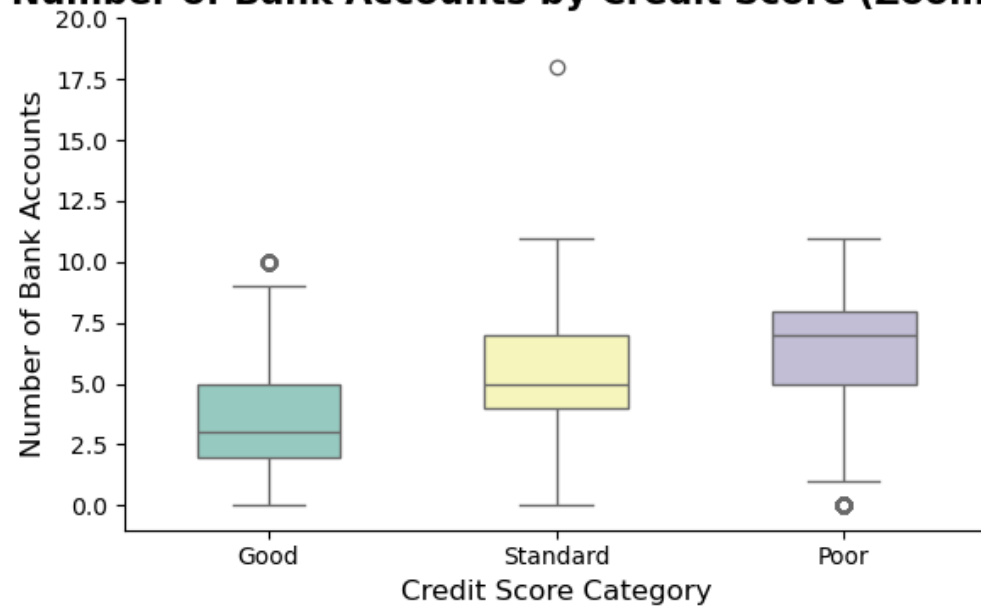
# 5 Payment Behaviour distribution by Credit Score
df_plot = df[df["Payment_Behaviour"] != "!@9#%8"].copy()

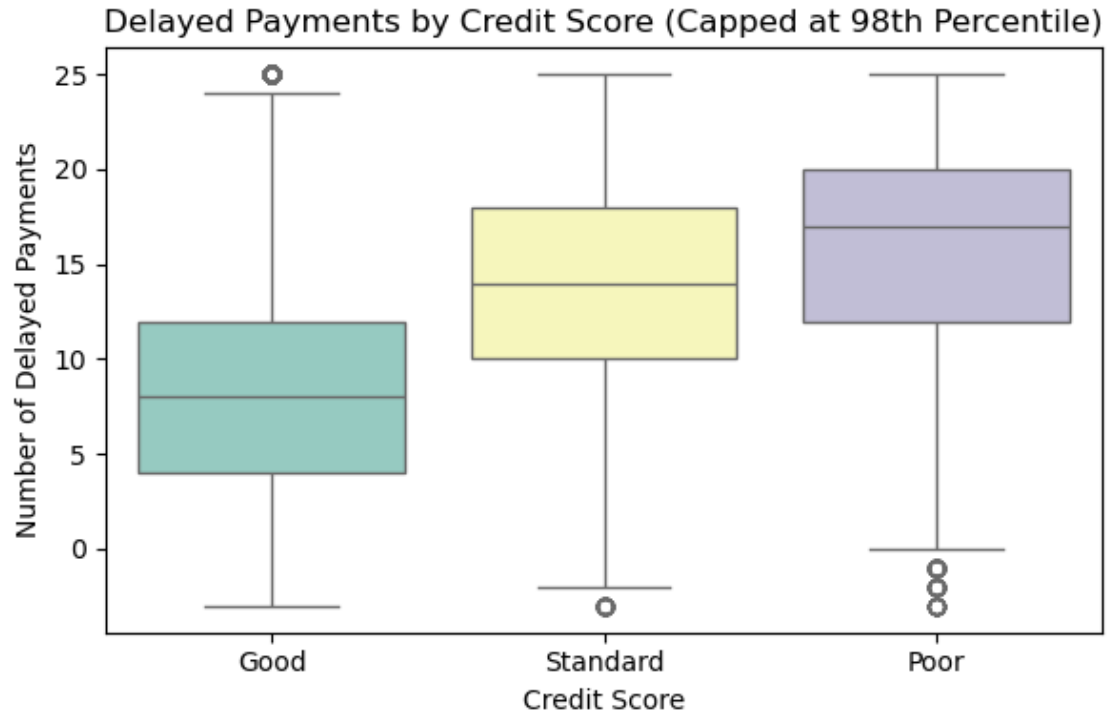
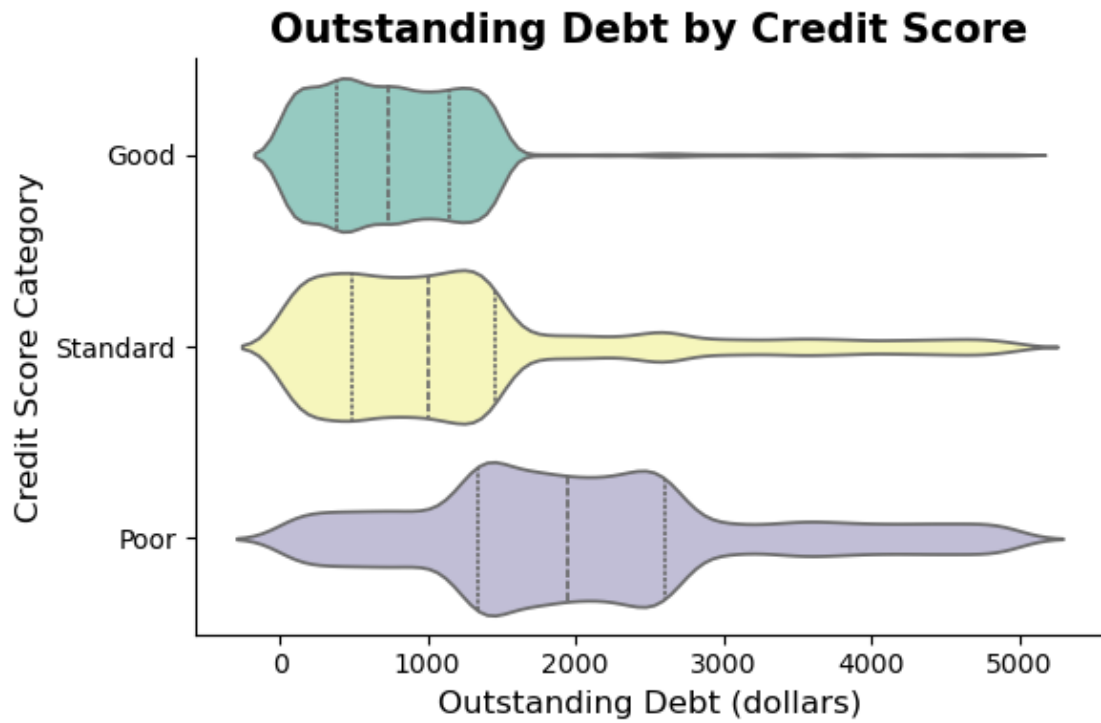
plt.figure(figsize=(10,4))
sns.countplot(
    data=df_plot,
    x="Payment_Behaviour",
    hue="Credit_Score",
    palette="Set3"
)
plt.xticks(rotation=45, ha="right")
plt.title("Payment Behaviour by Credit Score")
plt.tight_layout()
plt.show()

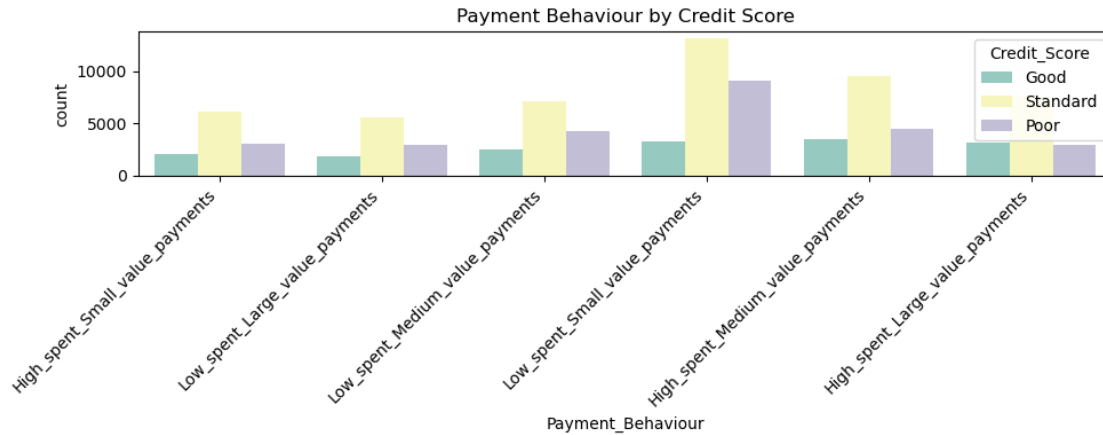
```



Number of Bank Accounts by Credit Score (Zoomed In)







```
[159]: # Correlation heatmap

# Create cleaned numeric copy
df_corr = df.copy()

# Define candidate numeric columns (from your list)
numeric_cols = [
    "Age",
    "Annual_Income",
    "Monthly_Inhand_Salary",
    "Num_Bank_Accounts",
    "Num_Credit_Card",
    "Interest_Rate",
    "Num_of_Loan",
    "Delay_from_due_date",
    "Num_of_Delayed_Payment",
    "Num_Credit_Inquiries",
    "Outstanding_Debt",
    "Credit_Utilization_Ratio",
    "Total_EMI_per_month",
    "Amount_invested_monthly",
    "Monthly_Balance"
]

# Fix dtype issues
for col in numeric_cols:
    df_corr[col] = (
        df_corr[col]
        .astype(str)
        .str.replace(",", "", regex=False)
```

```

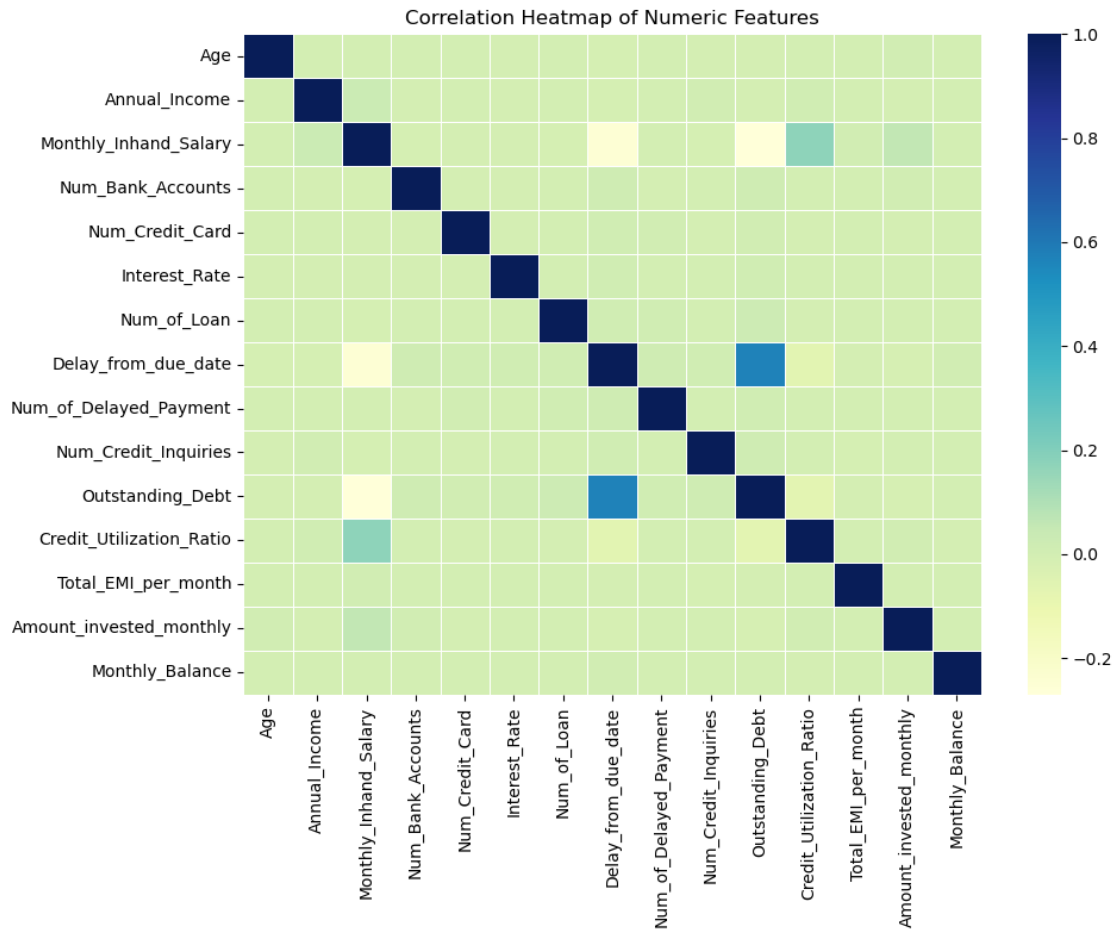
        .str.replace("_", "", regex=False)
    )
    df_corr[col] = pd.to_numeric(df_corr[col], errors="coerce")

# Handle invalid values
df_corr.loc[df_corr["Age"] < 0, "Age"] = np.nan
df_corr.loc[df_corr["Delay_from_due_date"] < 0, "Delay_from_due_date"] = np.nan

# Compute correlation matrix
corr_matrix = df_corr[numeric_cols].corr()

# Heatmap
plt.figure(figsize=(10, 8))
sns.heatmap(
    corr_matrix,
    cmap="YlGnBu",
    annot=False,
    linewidths=0.5
)
plt.title("Correlation Heatmap of Numeric Features")
plt.tight_layout()
plt.show()

```



```
[160]: # Data Cleaning
df_clean = df.copy()

def clean_numeric(s):
    return (
        s.astype(str)
        .str.replace(",", "", regex=False)
        .str.replace("_", "", regex=False)
        .replace(["NA", "NM", "None", ""], np.nan)
    )

numeric_object_cols = [
    "Age",
    "Annual_Income",
    "Num_of_Loan",
    "Num_of_Delayed_Payment",
    "Changed_Credit_Limit",
    "Outstanding_Debt",
```

```

    "Amount_invested_monthly",
]

for col in numeric_object_cols:
    df_clean[col] = clean_numeric(df_clean[col])
    df_clean[col] = pd.to_numeric(df_clean[col], errors="coerce")

# Age impossible (<0 or >100)
df_clean = df_clean[(df_clean["Age"] > 0) & (df_clean["Age"] < 100)]

# Num_of_Loan cannot be negative
df_clean = df_clean[df_clean["Num_of_Loan"] >= 0]

# Num_of_Delayed_Payment cannot be negative
df_clean = df_clean[(df_clean["Num_of_Delayed_Payment"] >= 0)]

# Outstanding_Debt cannot be negative
df_clean = df_clean[(df_clean["Outstanding_Debt"] >= 0)]

# Credit_Utilization_Ratio cannot be negative or above 100
# (if >100 it's fraud but not impossible, so only drop <0 here)
df_clean = df_clean[df_clean["Credit_Utilization_Ratio"] >= 0]

# Delay_from_due_date cannot be negative
df_clean = df_clean[df_clean["Delay_from_due_date"] >= 0]

# Interest rate cannot exceed 100% in any real consumer loan scenario
df_clean = df_clean[df_clean["Interest_Rate"] <= 100]

# Monthly_Balance: remove thousand separators and cast to numeric
df_clean["Monthly_Balance"] = (
    df_clean["Monthly_Balance"]
    .astype(str)
    .str.replace(",", "")
)
df_clean["Monthly_Balance"] = pd.to_numeric(df_clean["Monthly_Balance"],
errors="coerce")

# Parse Credit_History_Age: "XX Years and YY Months" -> total months
def parse_credit_age(x):
    if pd.isna(x):
        return np.nan
    parts = x.split(" ")
    years = int(parts[0])
    months = int(parts[3])
    return years * 12 + months

```



```
df_clean["Credit_History_Age"] = df_clean["Credit_History_Age"].
    ↪apply(parse_credit_age)

print("Shape before cleaning:", df.shape)
print("Shape after cleaning :", df_clean.shape)
print(df_clean.info())
print(df_clean.head())
print(df_clean.describe())
```

Shape before cleaning: (100000, 28)

Shape after cleaning : (84083, 28)

<class 'pandas.core.frame.DataFrame'>

Index: 84083 entries, 0 to 99999

Data columns (total 28 columns):

#	Column	Non-Null Count	Dtype
0	ID	84083 non-null	object
1	Customer_ID	84083 non-null	object
2	Month	84083 non-null	object
3	Name	75673 non-null	object
4	Age	84083 non-null	int64
5	SSN	84083 non-null	object
6	Occupation	84083 non-null	object
7	Annual_Income	84083 non-null	float64
8	Monthly_Inhand_Salary	71456 non-null	float64
9	Num_Bank_Accounts	84083 non-null	int64
10	Num_Credit_Card	84083 non-null	int64
11	Interest_Rate	84083 non-null	int64
12	Num_of_Loan	84083 non-null	int64
13	Type_of_Loan	74602 non-null	object
14	Delay_from_due_date	84083 non-null	int64
15	Num_of_Delayed_Payment	84083 non-null	float64
16	Changed_Credit_Limit	82323 non-null	float64
17	Num_Credit_Inquiries	82438 non-null	float64
18	Credit_Mix	84083 non-null	object
19	Outstanding_Debt	84083 non-null	float64
20	Credit_Utilization_Ratio	84083 non-null	float64
21	Credit_History_Age	76482 non-null	float64
22	Payment_of_Min_Amount	84083 non-null	object
23	Total_EMI_per_month	84083 non-null	float64
24	Amount_invested_monthly	80310 non-null	float64
25	Payment_Behaviour	84083 non-null	object
26	Monthly_Balance	83059 non-null	float64
27	Credit_Score	84083 non-null	object

dtypes: float64(11), int64(6), object(11)

memory usage: 18.6+ MB

None

ID Customer_ID Month Name Age SSN Occupation \

0	0x1602	CUS_0xd40	January	Aaron Maashoh	23	821-00-0265	Scientist
3	0x1605	CUS_0xd40	April	Aaron Maashoh	23	821-00-0265	Scientist
5	0x1607	CUS_0xd40	June	Aaron Maashoh	23	821-00-0265	Scientist
6	0x1608	CUS_0xd40	July	Aaron Maashoh	23	821-00-0265	Scientist
7	0x1609	CUS_0xd40	August	NaN	23	#F%\$D@*&8	Scientist

	Annual_Income	Monthly_Inhand_Salary	Num_Bank_Accounts	...	Credit_Mix	\
0	19114.12	1824.843333	3	...	-	
3	19114.12	NaN	3	...	Good	
5	19114.12	NaN	3	...	Good	
6	19114.12	1824.843333	3	...	Good	
7	19114.12	1824.843333	3	...	Good	

	Outstanding_Debt	Credit_Utilization_Ratio	Credit_History_Age	\
0	809.98	26.822620	265.0	
3	809.98	31.377862	268.0	
5	809.98	27.262259	270.0	
6	809.98	22.537593	271.0	
7	809.98	23.933795	NaN	

	Payment_of_Min_Amount	Total_EMI_per_month	Amount_invested_monthly	\
0	No	49.574949	80.415295	
3	No	49.574949	199.458074	
5	No	49.574949	62.430172	
6	No	49.574949	178.344067	
7	No	49.574949	24.785217	

	Payment_Behaviour	Monthly_Balance	Credit_Score
0	High_spent_Small_value_payments	312.494089	Good
3	Low_spent_Small_value_payments	223.451310	Good
5	!@9#%&8	340.479212	Good
6	Low_spent_Small_value_payments	244.565317	Good
7	High_spent_Medium_value_payments	358.124168	Standard

[5 rows x 28 columns]

	Age	Annual_Income	Monthly_Inhand_Salary	Num_Bank_Accounts	\
count	84083.000000	8.408300e+04	71456.000000	84083.000000	
mean	33.246542	1.756191e+05	4179.919725	17.126399	
std	10.771383	1.430040e+06	3173.471425	117.524251	
min	14.000000	7.005930e+03	303.645417	-1.000000	
25%	24.000000	1.940340e+04	1623.130833	4.000000	
50%	33.000000	3.742188e+04	3081.170833	6.000000	
75%	42.000000	7.255936e+04	5937.530000	8.000000	
max	99.000000	2.419806e+07	15204.633333	1798.000000	

	Num_Credit_Card	Interest_Rate	Num_of_Loan	Delay_from_due_date	\
count	84083.000000	84083.000000	84083.000000	84083.000000	
mean	22.528502	14.654009	7.159652	21.301452	

std	129.120822	8.805476	60.094392	14.787802
min	0.000000	1.000000	0.000000	0.000000
25%	4.000000	8.000000	2.000000	10.000000
50%	6.000000	13.000000	3.000000	18.000000
75%	7.000000	20.000000	5.000000	28.000000
max	1499.000000	100.000000	1496.000000	67.000000

	Num_of_Delayed_Payment	Changed_Credit_Limit	Num_Credit_Inquiries	\
count	84083.000000	82323.000000	82438.000000	
mean	31.181868	10.452756	27.609343	
std	226.093126	6.803577	191.817295	
min	0.000000	-6.480000	0.000000	
25%	9.000000	5.380000	3.000000	
50%	14.000000	9.470000	6.000000	
75%	18.000000	15.010000	9.000000	
max	4397.000000	36.490000	2597.000000	

	Outstanding_Debt	Credit_Utilization_Ratio	Credit_History_Age	\
count	84083.000000	84083.000000	76482.000000	
mean	1435.328080	32.296730	220.248987	
std	1159.127006	5.115194	99.708119	
min	0.230000	20.000000	1.000000	
25%	569.950000	28.073859	143.000000	
50%	1171.750000	32.322121	219.000000	
75%	1965.595000	36.510741	301.000000	
max	4998.070000	50.000000	404.000000	

	Total_EMI_per_month	Amount_invested_monthly	Monthly_Balance
count	84083.000000	80310.000000	83059.000000
mean	1411.335765	633.485496	401.304410
std	8318.777152	2035.524840	213.020056
min	0.000000	0.000000	0.007760
25%	30.506517	74.240839	269.796352
50%	69.272824	135.496347	335.991381
75%	161.499680	264.964253	468.343094
max	82331.000000	10000.000000	1602.040519

```
[161]: # Check whether the dataset satisfies I.I.D. assumptions before dropping
        ↪non-predictive ID columns

        # 1. Each Customer_ID should map to only one SSN
        cid_ssn_conflict = df_clean.groupby("Customer_ID")["SSN"].transform("nunique")
        ↪> 1

        # 2. Each Customer_ID should map to only one Name
        cid_name_conflict = df_clean.groupby("Customer_ID")["Name"].
        ↪transform("nunique") > 1
```

```

# 3. Each SSN should map to only one Customer_ID
ssn_cid_conflict = df_clean.groupby("SSN")["Customer_ID"].transform("nunique")
    ↪ > 1

# 4. Each SSN should map to only one Name
ssn_name_conflict = df_clean.groupby("SSN")["Name"].transform("nunique") > 1

# 5. ID should be unique, but check for safety
id_conflict = df_clean["ID"].duplicated(keep=False)

# Combine all conflict conditions
conflict_mask = (
    cid_ssn_conflict |
    cid_name_conflict |
    ssn_cid_conflict |
    ssn_name_conflict |
    id_conflict
)
# The misaligned rows (PS: NOT dropping them)
conflict_rows = df_clean[conflict_mask]
print("misaligned rows:", (conflict_rows.head(10)))

# Count unique valid person SSNs
persons = df_clean["SSN"].value_counts()
print("Number of unique persons:", len(persons))
print(persons.head())

"""Although each individual has 8 monthly records, this dataset is not treated
    ↪ as a classical time series problem. In a true time series task, the target
    ↪ at time t depends strictly on past values (e.g., ARIMA, forecasting). Here,
    ↪ the goal is classification of credit score, using all available features at
    ↪ each month. The temporal order does not define the predictive structure, and
    ↪ the target is not modeled as a sequence. Therefore, the dataset is best
    ↪ interpreted as panel data (repeated cross-sections) rather than a
    ↪ forecasting problem."""

# Drop non-predictive non-grouping columns
df_clean = df_clean.drop(columns=["Customer_ID", "Name", "ID"])

"""According to entity-resolution literature (Christen 2012; Wooldridge 2010),
    ↪ such inconsistencies in identity attributes are common in real-world and
    ↪ synthetic datasets and should not be used as a criterion for dropping
    ↪ observations, unless full entity resolution is performed.

```

Since the goal of this project is credit-score prediction-not identity,
 ↳reconciliation-these conflicts are treated as noisy metadata rather than,
 ↳erroneous samples. Dropping these rows would arbitrarily remove valid,
 ↳monthly observations and distort the distribution of the data.
 ""

misaligned rows:		ID	Customer_ID	Month	Name	Age
SSN	\					
0	0x1602	CUS_0xd40	January	Aaron Maashoh	23	821-00-0265
3	0x1605	CUS_0xd40	April	Aaron Maashoh	23	821-00-0265
5	0x1607	CUS_0xd40	June	Aaron Maashoh	23	821-00-0265
6	0x1608	CUS_0xd40	July	Aaron Maashoh	23	821-00-0265
7	0x1609	CUS_0xd40	August	NaN	23	#F%\$D@*&8
24	0x1626	CUS_0xb891	January	Jasond	54	072-31-6145
25	0x1627	CUS_0xb891	February	Jasond	54	072-31-6145
26	0x1628	CUS_0xb891	March	Jasond	55	072-31-6145
27	0x1629	CUS_0xb891	April	Jasond	55	072-31-6145
28	0x162a	CUS_0xb891	May	Jasond	55	072-31-6145

	Occupation	Annual_Income	Monthly_Inhand_Salary	Num_Bank_Accounts	\
0	Scientist	19114.12	1824.843333		3
3	Scientist	19114.12	NaN		3
5	Scientist	19114.12	NaN		3
6	Scientist	19114.12	1824.843333		3
7	Scientist	19114.12	1824.843333		3
24	Entrepreneur	30689.89	2612.490833		2
25	Entrepreneur	30689.89	2612.490833		2
26	Entrepreneur	30689.89	2612.490833		2
27	Entrepreneur	30689.89	2612.490833		2
28	Entrepreneur	30689.89	2612.490833		2

	...	Credit_Mix	Outstanding_Debt	Credit_Utilization_Ratio	\
0	...	-	809.98	26.822620	
3	...	Good	809.98	31.377862	
5	...	Good	809.98	27.262259	
6	...	Good	809.98	22.537593	
7	...	Good	809.98	23.933795	
24	...	Good	632.46	26.544229	
25	...	Good	632.46	35.279982	
26	...	Good	632.46	32.301163	
27	...	Good	632.46	38.132348	
28	...	Good	632.46	41.154317	

	Credit_History_Age	Payment_of_Min_Amount	Total_EMI_per_month	\
0	265.0	No	49.574949	
3	268.0	No	49.574949	
5	270.0	No	49.574949	
6	271.0	No	49.574949	

7	NaN	No	49.574949
24	207.0	No	16.415452
25	208.0	No	16.415452
26	209.0	NM	16.415452
27	210.0	No	16.415452
28	211.0	No	16.415452

	Amount_invested_monthly	Payment_Behaviour	Monthly_Balance \
0	80.415295	High_spent_Small_value_payments	312.494089
3	199.458074	Low_spent_Small_value_payments	223.451310
5	62.430172	!@9#%8	340.479212
6	178.344067	Low_spent_Small_value_payments	244.565317
7	24.785217	High_spent_Medium_value_payments	358.124168
24	81.228859	Low_spent_Large_value_payments	433.604773
25	124.881820	Low_spent_Small_value_payments	409.951812
26	83.406509	High_spent_Medium_value_payments	411.427123
27	272.334037	Low_spent_Small_value_payments	262.499594
28	10000.000000	Low_spent_Large_value_payments	359.374916

	Credit_Score
0	Good
3	Good
5	Good
6	Good
7	Standard
24	Standard
25	Standard
26	Standard
27	Standard
28	Standard

[10 rows x 28 columns]

Number of unique persons: 12501

SSN

#F%\$D@*&8	4665
939-69-9988	8
618-81-3518	8
888-44-5992	8
261-83-6127	8

Name: count, dtype: int64

[161]: 'According to entity-resolution literature (Christen 2012; Wooldridge 2010), such inconsistencies in identity attributes are common in real-world and synthetic datasets and should not be used as a criterion for dropping observations, unless full entity resolution is performed.\n\nSince the goal of this project is credit-score prediction-not identity reconciliation-these conflicts are treated as noisy metadata rather than erroneous samples. Dropping

these rows would arbitrarily remove valid monthly observations and distort the distribution of the data.\n'

```
[162]: # Check missing values for each column which are not written as NA
for col in df_clean.columns:
    uniques = df_clean[col].unique()
    print(f"\n--- {col} ---")
    print(f"Unique count: {len(uniques)}")
    # Only print first 20 to avoid flooding
    print(uniques[:20])
```

--- Month ---

Unique count: 8

['January' 'April' 'June' 'July' 'August' 'February' 'May' 'March']

--- Age ---

Unique count: 45

[23 28 34 54 55 21 31 33 30 24 44 40 41 35 36 39 37 20 45 46]

--- SSN ---

Unique count: 12501

['821-00-0265' '#F%\$D@*&8' '004-07-5839' '486-85-3974' '072-31-6145'
 '615-06-7821' '612-70-8987' '411-51-0676' '500-92-6408' '070-19-1622'
 '366-68-1681' '221-30-8554' '342-90-2649' '414-53-2918' '328-33-6328'
 '655-05-7666' '965-46-2491' '891-55-9364' '928-91-4452' '084-25-3745']

--- Occupation ---

Unique count: 16

['Scientist' '_____' 'Teacher' 'Engineer' 'Entrepreneur' 'Developer'
 'Lawyer' 'Media_Manager' 'Doctor' 'Journalist' 'Manager' 'Accountant'
 'Musician' 'Mechanic' 'Writer' 'Architect']

--- Annual_Income ---

Unique count: 13321

[1.9114120e+04 3.4847840e+04 1.4316264e+05 3.0689890e+04 3.5547710e+04
 7.3928460e+04 1.3131340e+05 1.0909427e+07 3.4081380e+04 1.1483841e+05
 3.1370800e+04 3.3751270e+04 8.8640240e+04 5.4392160e+04 8.7015450e+03
 2.5546260e+04 3.1993780e+04 9.2047080e+04 3.2284620e+04 9.7791420e+04]

--- Monthly_Inhand_Salary ---

Unique count: 13186

[1824.84333333 nan 3037.98666667 12187.22
 2612.49083333 2853.30916667 5988.705 11242.78333333
 10469.20775939 2611.115 9843.8675 2825.23333333
 2948.60583333 7266.68666667 4766.68 519.12875
 2415.855 2942.14833333 7591.59 2898.385]

--- Num_Bank_Accounts ---

Unique count: 832

```
[ 3  2  1  7  4  0  8  5  6  9 10 1414  67  572
 91 528 1647 1696 649 889]
```

--- Num_Credit_Card ---

Unique count: 1106

```
[ 4  5 1288  1  7  6 1029 488  8 1381 898  3 518 1005
 9 1189  2 10 514 943]
```

--- Interest_Rate ---

Unique count: 53

```
[ 3  6  8  4  5 15  7 12 20  1 14 32 16 17 10 31 25 18 19  9]
```

--- Num_of_Loan ---

Unique count: 371

```
[ 4  1  3 967  0  2  7  5  6  8  9 1464 622 352
1017 945 146 563 341 720]
```

--- Type_of_Loan ---

Unique count: 6261

```
['Auto Loan, Credit-Builder Loan, Personal Loan, and Home Equity Loan'
 'Credit-Builder Loan' 'Auto Loan, Auto Loan, and Not Specified'
 'Not Specified' nan 'Credit-Builder Loan, and Mortgage Loan'
 'Not Specified, Auto Loan, and Student Loan'
 'Personal Loan, Debt Consolidation Loan, and Auto Loan'
 'Not Specified, and Payday Loan'
 'Credit-Builder Loan, Personal Loan, and Auto Loan'
 'Payday Loan, and Payday Loan'
 'Not Specified, Student Loan, and Personal Loan'
 'Personal Loan, Payday Loan, Student Loan, Auto Loan, Home Equity Loan, Student
 Loan, and Payday Loan'
 'Not Specified, Student Loan, Student Loan, Credit-Builder Loan, and Auto Loan'
 'Payday Loan, and Home Equity Loan'
 'Credit-Builder Loan, Not Specified, Mortgage Loan, Payday Loan, Credit-Builder
 Loan, and Personal Loan'
 'Mortgage Loan, Debt Consolidation Loan, Payday Loan, Auto Loan, and Not
 Specified'
 'Credit-Builder Loan, Mortgage Loan, Mortgage Loan, Credit-Builder Loan, and
 Student Loan'
 'Not Specified, Student Loan, and Student Loan'
 'Payday Loan, Not Specified, Credit-Builder Loan, Debt Consolidation Loan,
 Payday Loan, Not Specified, Student Loan, and Student Loan']
```

--- Delay_from_due_date ---

Unique count: 68

```
[ 3  5  8  7 13 10  0  1  9 12 30 31 34 27 14 11  4  2 16 17]
```



```

--- Num_of_Delayed_Payment ---
Unique count: 650
[ 7.  4.  8.  6.  1.  3.  0.  5.  9. 15. 17. 10.  2. 14. 11. 20. 22. 13.
 16. 12.]

--- Changed_Credit_Limit ---
Unique count: 4278
[11.27  6.27  9.27  5.42  7.42  6.42  7.1  11.1  9.1  1.99 -2.01 -1.01
 -3.01  2.58 10.14  9.14  9.34  8.34 17.13 21.13]

--- Num_Credit_Inquiries ---
Unique count: 1074
[4.000e+00 2.000e+00 3.000e+00          nan 5.000e+00 9.000e+00 8.000e+00
 7.000e+00 6.000e+00 0.000e+00 1.000e+00 1.000e+01 1.050e+03 1.100e+01
 1.200e+01 1.044e+03 1.700e+01 1.936e+03 1.300e+01 5.680e+02]

--- Credit_Mix ---
Unique count: 4
['_' 'Good' 'Standard' 'Bad']

--- Outstanding_Debt ---
Unique count: 12203
[ 809.98  605.03 1303.01  632.46  943.86  548.2  352.16 1704.18 1377.74
 421.43 1328.93  950.36  179.22 2602.69  758.44  818.22 1296.64 1283.37
 107.41 2430.21]

--- Credit_Utilization_Ratio ---
Unique count: 84083
[26.82261962 31.37786187 27.26225871 22.53759303 23.9337948 24.46403064
 38.55084843 39.18265566 34.97789475 33.3810102 31.13170161 32.93385629
 28.61673482 41.70257342 26.51981539 39.50164811 31.37614957 39.78399346
 38.06862363 38.37475295]

--- Credit_History_Age ---
Unique count: 405
[265. 268. 270. 271.  nan 319. 320. 322. 323. 324. 325. 326. 213. 214.
 215. 217. 218. 219. 220. 207.]

--- Payment_of_Min_Amount ---
Unique count: 3
['No' 'NM' 'Yes']

--- Total_EMI_per_month ---
Unique count: 14453
[4.95749492e+01 1.88162146e+01 2.46992319e+02 1.64154517e+01
 0.00000000e+00 1.50150000e+04 1.55150000e+04 1.37644605e+02
 9.11220179e+02 2.38340000e+04 3.26620000e+04 7.04783327e+01
 2.26892792e+02 4.66161291e+01 1.64150000e+04 6.50081743e+01]

```

```
1.35173371e+02 8.03570000e+04 1.24392082e+02 3.65481972e+01]
```

```
--- Amount_invested_monthly ---
```

```
Unique count: 76582
```

```
[ 80.41529544 199.45807439 62.43017233 178.34406741
 24.78521651 104.29182517 40.39123783 99.30622796
130.11542024 43.47719014 70.10177421 218.90434353
168.41370268 232.86038376 10000. 825.21626994
430.94752788 257.80809943 263.17416316 81.22885871]
```

```
--- Payment_Behaviour ---
```

```
Unique count: 7
```

```
['High_spent_Small_value_payments' 'Low_spent_Small_value_payments'
 '!@9#%8' 'High_spent_Medium_value_payments'
 'High_spent_Large_value_payments' 'Low_spent_Medium_value_payments'
 'Low_spent_Large_value_payments']
```

```
--- Monthly_Balance ---
```

```
Unique count: 83060
```

```
[ 312.49408868 223.45130973 340.47921179 244.56531671 358.12416761
 470.69062693 484.59121427 465.67622413 444.86703185 481.50526195
 464.88067789 356.07810856 1043.31597787 998.86929679 715.7413674
 426.51341061 810.78215267 963.92158112 968.55551738 895.49458318]
```

```
--- Credit_Score ---
```

```
Unique count: 3
```

```
['Good' 'Standard' 'Poor']
```

```
[163]: # Handle special missing value tokens in categorical columns
missing_tokens = ["_", "_____", "!@9#%8", "NM"]

for col in ["Occupation", "Credit_Mix", "Payment_Behaviour",
            ↪ "Payment_of_Min_Amount"]:
    df_clean[col] = df_clean[col].replace(missing_tokens, np.nan)
```

```
[164]: # Re-run the EDA visualizations on clean_df as needed
# Missing values per column in clean_df
missing_counts = df_clean.isnull().sum().sort_values(ascending=False)

# Columns to exclude
exclude_cols = ["SSN"]

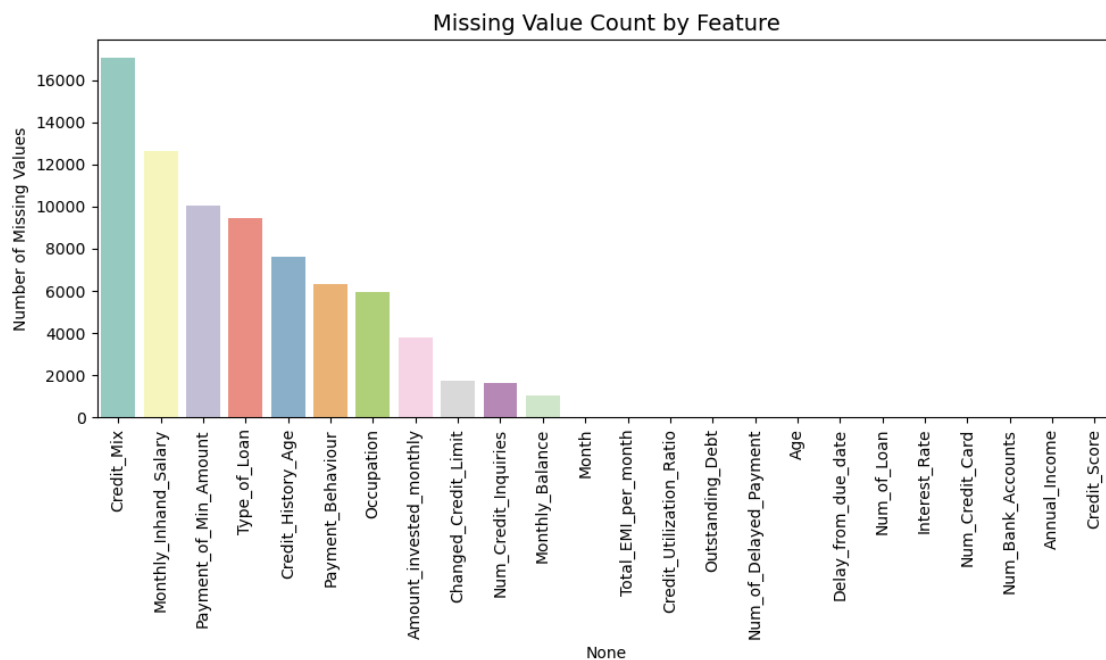
# Remove excluded columns
missing_counts = missing_counts.drop(labels=[c for c in exclude_cols if c in
            ↪ missing_counts.index])

plt.figure(figsize=(10, 6))
```

```

sns.barplot(
    x=missing_counts.index,
    y=missing_counts.values,
    hue=missing_counts.index,
    palette="Set3"
)
plt.title("Missing Value Count by Feature", fontsize=14)
plt.ylabel("Number of Missing Values")
plt.xticks(rotation=90)
plt.tight_layout()
plt.show()

```



```

[165]: # Re-run some of the previous EDA plots on clean_df if needed
# The distribution of Credit_Score after cleaning
plt.figure(figsize=(6,4))
sns.countplot(data=df_clean, x="Credit_Score", hue="Credit_Score",
    palette="Set3")
plt.title("Distribution of Credit Score Classes")
plt.xlabel("Credit Score")
plt.ylabel("Count")
plt.tight_layout()
plt.show()

# Heatmap of correlations in clean_df
numeric_cols = [

```

```

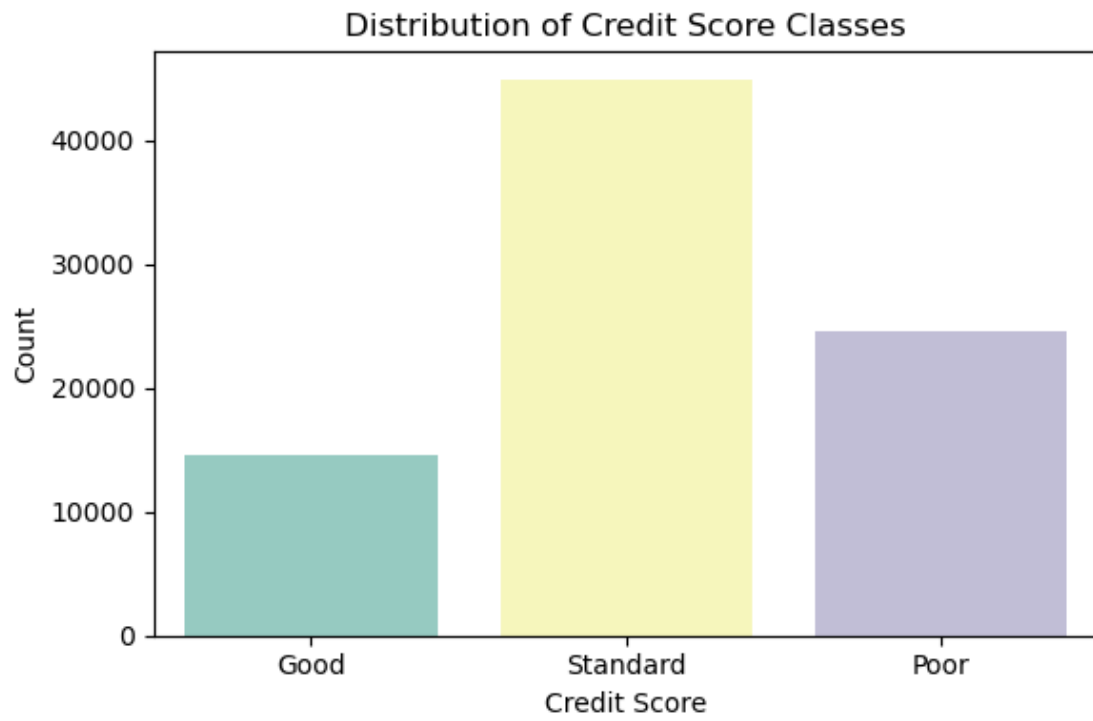
    "Age", "Annual_Income", "Monthly_Inhand_Salary", "Num_Bank_Accounts",
    "Num_Credit_Card", "Interest_Rate", "Num_of_Loan", "Delay_from_due_date",
    "Num_of_Delayed_Payment", "Num_Credit_Inquiries", "Outstanding_Debt",
    "Credit_Utilization_Ratio", "Total_EMI_per_month",
    "Amount_invested_monthly", "Monthly_Balance"
]

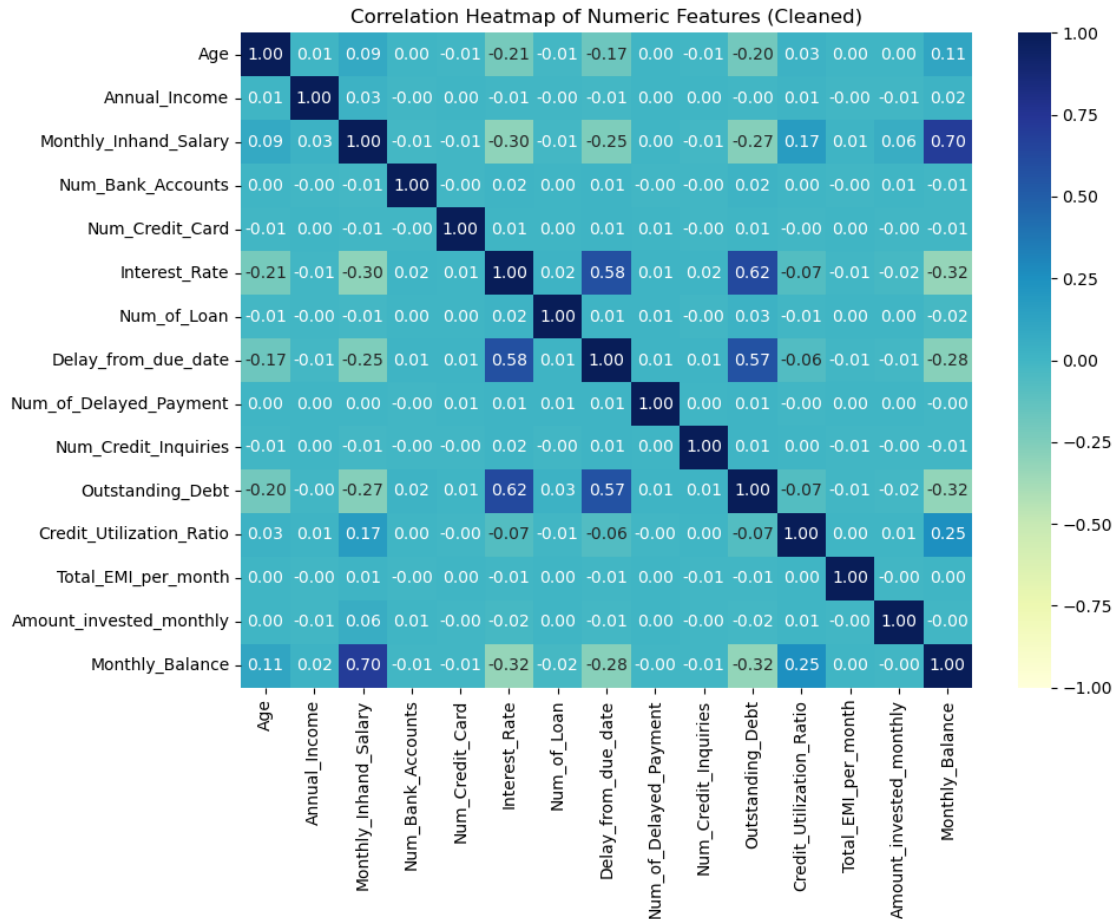
corr = df_clean[numeric_cols].corr()

plt.figure(figsize=(10, 8))
sns.heatmap(
    corr,
    cmap="YlGnBu",
    vmin=-1, vmax=1,
    annot=True, fmt=".2f"
)
plt.title("Correlation Heatmap of Numeric Features (Cleaned)")
plt.tight_layout()
plt.show()

print(df_clean.info())
for col in df_clean.columns:
    uniques = df_clean[col].unique()
    print(f"\n--- {col} ---")
    print(f"Unique count: {len(uniques)}")
    # Only print first 20 to avoid flooding
    print(uniques[:20])

```





```
<class 'pandas.core.frame.DataFrame'>
```

```
Index: 84083 entries, 0 to 99999
```

```
Data columns (total 25 columns):
```

#	Column	Non-Null Count	Dtype
0	Month	84083 non-null	object
1	Age	84083 non-null	int64
2	SSN	84083 non-null	object
3	Occupation	78153 non-null	object
4	Annual_Income	84083 non-null	float64
5	Monthly_Inhand_Salary	71456 non-null	float64
6	Num_Bank_Accounts	84083 non-null	int64
7	Num_Credit_Card	84083 non-null	int64
8	Interest_Rate	84083 non-null	int64
9	Num_of_Loan	84083 non-null	int64
10	Type_of_Loan	74602 non-null	object
11	Delay_from_due_date	84083 non-null	int64
12	Num_of_Delayed_Payment	84083 non-null	float64
13	Changed_Credit_Limit	82323 non-null	float64

```

14 Num_Credit_Inquiries      82438 non-null float64
15 Credit_Mix                67024 non-null object
16 Outstanding_Debt          84083 non-null float64
17 Credit_Utilization_Ratio  84083 non-null float64
18 Credit_History_Age        76482 non-null float64
19 Payment_of_Min_Amount     74032 non-null object
20 Total_EMI_per_month       84083 non-null float64
21 Amount_invested_monthly   80310 non-null float64
22 Payment_Behaviour         77757 non-null object
23 Monthly_Balance           83059 non-null float64
24 Credit_Score              84083 non-null object
dtypes: float64(11), int64(6), object(8)
memory usage: 16.7+ MB
None

--- Month ---
Unique count: 8
['January' 'April' 'June' 'July' 'August' 'February' 'May' 'March']

--- Age ---
Unique count: 45
[23 28 34 54 55 21 31 33 30 24 44 40 41 35 36 39 37 20 45 46]

--- SSN ---
Unique count: 12501
['821-00-0265' '#F%$D@*&8' '004-07-5839' '486-85-3974' '072-31-6145'
'615-06-7821' '612-70-8987' '411-51-0676' '500-92-6408' '070-19-1622'
'366-68-1681' '221-30-8554' '342-90-2649' '414-53-2918' '328-33-6328'
'655-05-7666' '965-46-2491' '891-55-9364' '928-91-4452' '084-25-3745']

--- Occupation ---
Unique count: 16
['Scientist' nan 'Teacher' 'Engineer' 'Entrepreneur' 'Developer' 'Lawyer'
'Media_Manager' 'Doctor' 'Journalist' 'Manager' 'Accountant' 'Musician'
'Mechanic' 'Writer' 'Architect']

--- Annual_Income ---
Unique count: 13321
[1.9114120e+04 3.4847840e+04 1.4316264e+05 3.0689890e+04 3.5547710e+04
7.3928460e+04 1.3131340e+05 1.0909427e+07 3.4081380e+04 1.1483841e+05
3.1370800e+04 3.3751270e+04 8.8640240e+04 5.4392160e+04 8.7015450e+03
2.5546260e+04 3.1993780e+04 9.2047080e+04 3.2284620e+04 9.7791420e+04]

--- Monthly_Inhand_Salary ---
Unique count: 13186
[ 1824.84333333          nan  3037.98666667 12187.22
  2612.49083333  2853.30916667  5988.705      11242.78333333
 10469.20775939  2611.115      9843.8675      2825.23333333]

```

```

2948.60583333 7266.68666667 4766.68      519.12875
2415.855      2942.14833333 7591.59      2898.385      ]

```

--- Num_Bank_Accounts ---

Unique count: 832

```

[ 3  2  1  7  4  0  8  5  6  9  10 1414  67  572
 91 528 1647 1696 649 889]

```

--- Num_Credit_Card ---

Unique count: 1106

```

[ 4  5 1288  1  7  6 1029 488  8 1381 898  3 518 1005
 9 1189  2 10 514 943]

```

--- Interest_Rate ---

Unique count: 53

```

[ 3 6 8 4 5 15 7 12 20 1 14 32 16 17 10 31 25 18 19 9]

```

--- Num_of_Loan ---

Unique count: 371

```

[ 4  1  3 967  0  2  7  5  6  8  9 1464 622 352
1017 945 146 563 341 720]

```

--- Type_of_Loan ---

Unique count: 6261

```

['Auto Loan, Credit-Builder Loan, Personal Loan, and Home Equity Loan'
 'Credit-Builder Loan' 'Auto Loan, Auto Loan, and Not Specified'
 'Not Specified' nan 'Credit-Builder Loan, and Mortgage Loan'
 'Not Specified, Auto Loan, and Student Loan'
 'Personal Loan, Debt Consolidation Loan, and Auto Loan'
 'Not Specified, and Payday Loan'
 'Credit-Builder Loan, Personal Loan, and Auto Loan'
 'Payday Loan, and Payday Loan'
 'Not Specified, Student Loan, and Personal Loan'
 'Personal Loan, Payday Loan, Student Loan, Auto Loan, Home Equity Loan, Student
Loan, and Payday Loan'
 'Not Specified, Student Loan, Student Loan, Credit-Builder Loan, and Auto Loan'
 'Payday Loan, and Home Equity Loan'
 'Credit-Builder Loan, Not Specified, Mortgage Loan, Payday Loan, Credit-Builder
Loan, and Personal Loan'
 'Mortgage Loan, Debt Consolidation Loan, Payday Loan, Auto Loan, and Not
Specified'
 'Credit-Builder Loan, Mortgage Loan, Mortgage Loan, Credit-Builder Loan, and
Student Loan'
 'Not Specified, Student Loan, and Student Loan'
 'Payday Loan, Not Specified, Credit-Builder Loan, Debt Consolidation Loan,
Payday Loan, Not Specified, Student Loan, and Student Loan']

```

--- Delay_from_due_date ---


```

Unique count: 68
[ 3  5  8  7 13 10  0  1  9 12 30 31 34 27 14 11  4  2 16 17]

--- Num_of_Delayed_Payment ---
Unique count: 650
[ 7.  4.  8.  6.  1.  3.  0.  5.  9. 15. 17. 10.  2. 14. 11. 20. 22. 13.
 16. 12.]

--- Changed_Credit_Limit ---
Unique count: 4278
[11.27  6.27  9.27  5.42  7.42  6.42  7.1  11.1  9.1  1.99 -2.01 -1.01
 -3.01  2.58 10.14  9.14  9.34  8.34 17.13 21.13]

--- Num_Credit_Inquiries ---
Unique count: 1074
[4.000e+00 2.000e+00 3.000e+00      nan 5.000e+00 9.000e+00 8.000e+00
 7.000e+00 6.000e+00 0.000e+00 1.000e+00 1.000e+01 1.050e+03 1.100e+01
 1.200e+01 1.044e+03 1.700e+01 1.936e+03 1.300e+01 5.680e+02]

--- Credit_Mix ---
Unique count: 4
[nan 'Good' 'Standard' 'Bad']

--- Outstanding_Debt ---
Unique count: 12203
[ 809.98  605.03 1303.01  632.46  943.86  548.2  352.16 1704.18 1377.74
  421.43 1328.93  950.36  179.22 2602.69  758.44  818.22 1296.64 1283.37
 107.41 2430.21]

--- Credit_Utilization_Ratio ---
Unique count: 84083
[26.82261962 31.37786187 27.26225871 22.53759303 23.9337948  24.46403064
 38.55084843 39.18265566 34.97789475 33.3810102  31.13170161 32.93385629
 28.61673482 41.70257342 26.51981539 39.50164811 31.37614957 39.78399346
 38.06862363 38.37475295]

--- Credit_History_Age ---
Unique count: 405
[265. 268. 270. 271.  nan 319. 320. 322. 323. 324. 325. 326. 213. 214.
 215. 217. 218. 219. 220. 207.]

--- Payment_of_Min_Amount ---
Unique count: 3
['No' nan 'Yes']

--- Total_EMI_per_month ---
Unique count: 14453
[4.95749492e+01 1.88162146e+01 2.46992319e+02 1.64154517e+01

```

```
0.00000000e+00 1.50150000e+04 1.55150000e+04 1.37644605e+02
9.11220179e+02 2.38340000e+04 3.26620000e+04 7.04783327e+01
2.26892792e+02 4.66161291e+01 1.64150000e+04 6.50081743e+01
1.35173371e+02 8.03570000e+04 1.24392082e+02 3.65481972e+01]
```

```
--- Amount_invested_monthly ---
```

```
Unique count: 76582
```

```
[ 80.41529544 199.45807439 62.43017233 178.34406741
 24.78521651 104.29182517 40.39123783 99.30622796
130.11542024 43.47719014 70.10177421 218.90434353
168.41370268 232.86038376 10000. 825.21626994
430.94752788 257.80809943 263.17416316 81.22885871]
```

```
--- Payment_Behaviour ---
```

```
Unique count: 7
```

```
['High_spent_Small_value_payments' 'Low_spent_Small_value_payments' nan
'High_spent_Medium_value_payments' 'High_spent_Large_value_payments'
'Low_spent_Medium_value_payments' 'Low_spent_Large_value_payments']
```

```
--- Monthly_Balance ---
```

```
Unique count: 83060
```

```
[ 312.49408868 223.45130973 340.47921179 244.56531671 358.12416761
 470.69062693 484.59121427 465.67622413 444.86703185 481.50526195
 464.88067789 356.07810856 1043.31597787 998.86929679 715.7413674
 426.51341061 810.78215267 963.92158112 968.55551738 895.49458318]
```

```
--- Credit_Score ---
```

```
Unique count: 3
```

```
['Good' 'Standard' 'Poor']
```

```
[166]: # Type of loan is nested categorical, so we will manually encode it
# Fill missing with "Missing"
df_clean["Type_of_Loan"] = df_clean["Type_of_Loan"].fillna("Missing")

# Extract unique loan types
def split_loan(text):
    text = str(text).replace("and", "").replace(" ", " ")
    parts = [p.strip() for p in text.split(",") if p.strip() != ""]
    return parts if parts else ["Missing"]

unique_loans = set()
for row in df_clean["Type_of_Loan"]:
    for item in split_loan(row):
        unique_loans.add(item)

unique_loans = sorted(list(unique_loans))
print("Loan types found:", unique_loans)
```

```

# Create binary columns for each loan type and drop original column
for loan in unique_loans:
    df_clean[f"Loan_{loan}"] = df_clean["Type_of_Loan"].apply(
        lambda x: 1 if loan in split_loan(x) else 0
    )
df_clean = df_clean.drop(columns=["Type_of_Loan"])

# Data Splitting and Preprocessing
X = df_clean.drop(columns="Credit_Score")
y = df_clean["Credit_Score"]
groups = df_clean["SSN"]

# GroupShuffleSplit
from sklearn.model_selection import GroupShuffleSplit

gss = GroupShuffleSplit(n_splits=1, test_size=0.15, random_state=42)
train_val_idx, test_idx = next(gss.split(X, y, groups))

X_train_val = X.iloc[train_val_idx].copy()
X_test = X.iloc[test_idx].copy()
y_train_val = y.iloc[train_val_idx].copy()
y_test = y.iloc[test_idx].copy()

X_test = X_test.drop(columns=["SSN"])

# Store the test set for final evaluation later
test_set = X_test.copy()
test_set["Credit_Score"] = y_test.values

test_set.to_csv("../data/test.csv", index=False)
print("Saved to data/test.csv")

```

Loan types found: ['Auto Loan', 'Credit-Builder Loan', 'Debt Consolidation Loan', 'Home Equity Loan', 'Missing', 'Mortgage Loan', 'Not Specified', 'Payday Loan', 'Personal Loan', 'Student Loan']
 Saved to data/test.csv

```

[167]: # Preprocessing pipeline (Handle missing values, categorical encoding, scaling,
      ↪ etc.)
# Build the column transformer
from sklearn.experimental import enable_iterative_imputer

from sklearn.compose import ColumnTransformer
from sklearn.pipeline import Pipeline
from sklearn.impute import SimpleImputer
from sklearn.preprocessing import OrdinalEncoder

```

```

from sklearn.impute import IterativeImputer
from sklearn.preprocessing import StandardScaler, OneHotEncoder

# Identify ordinal and categorical columns

ordinal_cols = ["Credit_Mix", "Payment_of_Min_Amount"]
Credit_Mix_order = ["Missing", "Bad", "Standard", "Good"]
Payment_of_Min_Amount_order = ["Missing", "Yes", "No"]

categorical_cols = ["Month", "Occupation", "Payment_Behaviour"]

numeric_cols = ["Age", "Annual_Income", "Monthly_Inhand_Salary",
↳ "Num_Bank_Accounts", "Num_Credit_Card", "Interest_Rate", "Num_of_Loan",
↳ "Delay_from_due_date", "Num_of_Delayed_Payment", "Changed_Credit_Limit",
↳ "Num_Credit_Inquiries", "Outstanding_Debt",
↳ "Credit_Utilization_Ratio", "Credit_History_Age", "Total_EMI_per_month",
↳ "Amount_invested_monthly", "Monthly_Balance"]

categorical_pipeline = Pipeline([
    ("imputer", SimpleImputer(strategy="constant", fill_value="Missing")),
    ("onehot", OneHotEncoder(handle_unknown="ignore"))
])

ordinal_pipeline = Pipeline([
    ("imputer", SimpleImputer(strategy="constant", fill_value="Missing")),
    ("encoder", OrdinalEncoder(categories=[Credit_Mix_order,
↳ Payment_of_Min_Amount_order]))
])

numeric_pipeline = Pipeline([
    ("passthrough", "passthrough")
])

preprocess = ColumnTransformer(
    transformers=[
        ("cat", categorical_pipeline, categorical_cols),
        ("ord", ordinal_pipeline, ordinal_cols),
        ("num", numeric_pipeline, numeric_cols)
    ]
)

full_preprocess = Pipeline([
    ("prep", preprocess),
    ("impute", IterativeImputer(random_state=i)),
    ("scale", StandardScaler())
])

```

```

# Define a K-fold cross-validation scheme
from sklearn.model_selection import GroupKFold

groups = X_train_val["SSN"]
X_train_val = X_train_val.drop(columns=["SSN"])

gkf = GroupKFold(n_splits=5)

```

```

[214]: # Hyperparameter tuning for Logistic Regression
from sklearn.model_selection import GridSearchCV
from sklearn.linear_model import LogisticRegression
from sklearn.pipeline import Pipeline

logreg_scores = []

# Put model in a pipeline with preprocessing
for i in range(5):
    print(f"Round {i+1}: ")
    logreg_pipeline = Pipeline([
        ("prep", full_preprocess),
        ("model", LogisticRegression(
            solver="saga",
            penalty="l2",
            max_iter=8000,
            n_jobs=-1
        ))
    ])

    # Define hyperparameter grid
    param_grid_logreg = {
        "model__C": [0.01, 0.1, 1, 10],
    }

    # Set up GridSearchCV
    logreg_grid = GridSearchCV(
        estimator=logreg_pipeline,
        param_grid=param_grid_logreg,
        cv=gkf,
        scoring="f1",
        n_jobs=-1
    )

    logreg_grid.fit(X_train_val, y_train_val, groups=groups)

    logreg_scores.append(logreg_grid.best_score_)

    print("Best params:", logreg_grid.best_params_)

```

```
print("Best CV score:", logreg_grid.best_score_)
```

Round 1:

```
/opt/anaconda3/lib/python3.13/site-  
packages/sklearn/model_selection/_validation.py:960: UserWarning: Scoring  
failed. The score on this train-test partition for these parameters will be set  
to nan. Details:
```

Traceback (most recent call last):

```
File "/opt/anaconda3/lib/python3.13/site-  
packages/sklearn/model_selection/_validation.py", line 949, in _score  
    scores = scorer(estimator, X_test, y_test, **score_params)  
File "/opt/anaconda3/lib/python3.13/site-packages/sklearn/metrics/_scorer.py",  
line 288, in __call__  
    return self._score(partial(_cached_call, None), estimator, X, y_true,  
    **_kwargs)
```

```
~~~~~  
File "/opt/anaconda3/lib/python3.13/site-packages/sklearn/metrics/_scorer.py",  
line 380, in _score  
    y_pred = method_caller(  
        estimator,  
        ...<2 lines>...  
        pos_label=pos_label,  
    )
```

```
File "/opt/anaconda3/lib/python3.13/site-packages/sklearn/metrics/_scorer.py",  
line 90, in _cached_call  
    result, _ = _get_response_values(  
        ~~~~~~  
        estimator, *args, response_method=response_method, **kwargs  
        ~~~~~~  
    )  
    ~
```

```
File "/opt/anaconda3/lib/python3.13/site-packages/sklearn/utils/_response.py",  
line 207, in _get_response_values  
    raise ValueError(  
        ...<2 lines>...  
    )
```

```
ValueError: pos_label=1 is not a valid label: It should be one of ['Good' 'Poor'  
'Standard']
```

```
warnings.warn(  
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/opt/anaconda3/lib/python3.13/site-
packages/sklearn/model_selection/_search.py:1108: UserWarning: One or more of
the test scores are non-finite: [nan nan nan nan]
    warnings.warn(

Best params: {'model__C': 0.01}
Best CV score: nan
Round 2:

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/opt/anaconda3/lib/python3.13/site-  
packages/sklearn/model_selection/_search.py:1108: UserWarning: One or more of  
the test scores are non-finite: [nan nan nan nan]
```

```
warnings.warn(  
  
Best params: {'model__C': 0.01}  
Best CV score: nan  
Round 3:
```

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/opt/anaconda3/lib/python3.13/site-
packages/sklearn/model_selection/_search.py:1108: UserWarning: One or more of
the test scores are non-finite: [nan nan nan nan]
    warnings.warn(

Best params: {'model__C': 0.01}
Best CV score: nan
Round 4:

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/opt/anaconda3/lib/python3.13/site-
packages/sklearn/model_selection/_search.py:1108: UserWarning: One or more of
the test scores are non-finite: [nan nan nan nan]
    warnings.warn(

Best params: {'model__C': 0.01}
Best CV score: nan
Round 5:

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**_kwargs)
~~~~~^~~~~~
File "/opt/anaconda3/lib/python3.13/site-packages/sklearn/metrics/_scorer.py",
line 380, in _score
    y_pred = method_caller(
        estimator,
    ...<2 lines>...
        pos_label=pos_label,
    )
File "/opt/anaconda3/lib/python3.13/site-packages/sklearn/metrics/_scorer.py",
line 90, in _cached_call
    result, _ = _get_response_values(
        ~~~~~^
        estimator, *args, response_method=response_method, **kwargs
    )
    ~
File "/opt/anaconda3/lib/python3.13/site-packages/sklearn/utils/_response.py",
line 207, in _get_response_values
    raise ValueError(
    ...<2 lines>...
    )
ValueError: pos_label=1 is not a valid label: It should be one of ['Good' 'Poor'
'Standard']

warnings.warn(

```

```

-----
KeyboardInterrupt                                Traceback (most recent call last)
Cell In[214], line 35
    26 # Set up GridSearchCV
    27 logreg_grid = GridSearchCV(
    28     estimator=logreg_pipeline,
    29     param_grid=param_grid_logreg,
    (...)
    32     n_jobs=-1
    33 )
--> 35 logreg_grid.fit(X_train_val, y_train_val, groups=groups)
    37 logreg_scores.append(logreg_grid.best_score_)
    39 print("Best params:", logreg_grid.best_params_)

File /opt/anaconda3/lib/python3.13/site-packages/sklearn/base.py:1389, in
↳ _fit_context.<locals>.decorator.<locals>.wrapper(estimator, *args, **kwargs)
    1382     estimator._validate_params()
    1384 with config_context(

```

```

1385     skip_parameter_validation=(
1386         prefer_skip_nested_validation or global_skip_validation
1387     )
1388 ):
-> 1389     return fit_method(estimator, *args, **kwargs)

File /opt/anaconda3/lib/python3.13/site-packages/sklearn/model_selection/_search.py:
py:1024, in BaseSearchCV.fit(self, X, y, **params)
    1018     results = self._format_results(
    1019         all_candidate_params, n_splits, all_out, all_more_results
    1020     )
    1022     return results
-> 1024 self._run_search(evaluate_candidates)
    1026 # multimetric is determined here because in the case of a callable
    1027 # self.scoring the return type is only known after calling
    1028 first_test_score = all_out[0]["test_scores"]

File /opt/anaconda3/lib/python3.13/site-packages/sklearn/model_selection/_search.py:
py:1571, in GridSearchCV._run_search(self, evaluate_candidates)
    1569 def _run_search(self, evaluate_candidates):
    1570     """Search all candidates in param_grid"""
-> 1571     evaluate_candidates(ParameterGrid(self.param_grid))

File /opt/anaconda3/lib/python3.13/site-packages/sklearn/model_selection/_search.py:
py:970, in BaseSearchCV.fit.<locals>.evaluate_candidates(candidate_params, cv,
more_results)
    962 if self.verbose > 0:
    963     print(
    964         "Fitting {0} folds for each of {1} candidates,"
    965         " totalling {2} fits".format(
    966             n_splits, n_candidates, n_candidates * n_splits
    967         )
    968     )
--> 970 out = parallel(
    971     delayed(_fit_and_score)(
    972         clone(base_estimator),
    973         X,
    974         y,
    975         train=train,
    976         test=test,
    977         parameters=parameters,
    978         split_progress=(split_idx, n_splits),
    979         candidate_progress=(cand_idx, n_candidates),
    980         **fit_and_score_kwargs,
    981     )
    982     for (cand_idx, parameters), (split_idx, (train, test)) in product(
    983         enumerate(candidate_params),
    984         enumerate(cv.split(X, y, **routed_params.splitter.split)),

```

```

985     )
986 )
988 if len(out) < 1:
989     raise ValueError(
990         "No fits were performed. "
991         "Was the CV iterator empty? "
992         "Were there no candidates?"
993     )

```

File /opt/anaconda3/lib/python3.13/site-packages/sklearn/utils/parallel.py:77, in

```

↳ in Parallel.__call__(self, iterable)
    72 config = get_config()
    73 iterable_with_config = (
    74     (_with_config(delayed_func, config), args, kwargs)
    75     for delayed_func, args, kwargs in iterable
    76 )
--> 77 return super().__call__(iterable_with_config)

```

File /opt/anaconda3/lib/python3.13/site-packages/joblib/parallel.py:2007, in

```

↳ Parallel.__call__(self, iterable)
    2001 # The first item from the output is blank, but it makes the interpreter
    2002 # progress until it enters the Try/Except block of the generator and
    2003 # reaches the first `yield` statement. This starts the asynchronous
    2004 # dispatch of the tasks to the workers.
    2005 next(output)
-> 2007 return output if self.return_generator else list(output)

```

File /opt/anaconda3/lib/python3.13/site-packages/joblib/parallel.py:1650, in

```

↳ Parallel._get_outputs(self, iterator, pre_dispatch)
    1647     yield
    1649     with self._backend.retrieval_context():
-> 1650         yield from self._retrieve()
    1652 except GeneratorExit:
    1653     # The generator has been garbage collected before being fully
    1654     # consumed. This aborts the remaining tasks if possible and warn
    1655     # the user if necessary.
    1656     self._exception = True

```

File /opt/anaconda3/lib/python3.13/site-packages/joblib/parallel.py:1762, in

```

↳ Parallel._retrieve(self)
    1757 # If the next job is not ready for retrieval yet, we just wait for
    1758 # async callbacks to progress.
    1759 if ((len(self._jobs) == 0) or
    1760     (self._jobs[0].get_status(
    1761         timeout=self.timeout) == TASK_PENDING)):
-> 1762     time.sleep(0.01)
    1763     continue
    1765 # We need to be careful: the job list can be filling up as

```

```
1766 # we empty it and Python list are not thread-safe by
1767 # default hence the use of the lock
```

KeyboardInterrupt:

```
[ ]: #Hyperparameter tuning for Linear SVM (LinearSVC)
from sklearn.linear_model import SGDClassifier

linear_svc_scores = []

for i in range():
    print(f"\nRound {i+1}: ")

    svc_pipeline = Pipeline([
        ("prep", full_preprocess),
        ("model", SGDClassifier(
            loss="hinge",
            penalty="l2",
            max_iter=2000,
            n_jobs=-1,
            tol=1e-3
        ))
    ])

    param_grid_svc = {
        "model__alpha": [1e-6, 1e-5, 1e-4]
    }

    svc_grid = GridSearchCV(
        estimator=svc_pipeline,
        param_grid=param_grid_svc,
        cv=gkf,
        scoring="f1",
        n_jobs=-1
    )

    svc_grid.fit(X_train_val, y_train_val, groups=groups)

    linear_svc_scores.append(svc_grid.best_score_)

    print("Best params:", svc_grid.best_params_)
    print("Best CV score:", svc_grid.best_score_)
```

Round 1:

Best params: {'model__alpha': 1e-05}

Best CV score: 0.5260185334757543

Round 2:
Best params: {'model__alpha': 1e-05}
Best CV score: 0.4990849641196961

Round 3:
Best params: {'model__alpha': 1e-05}
Best CV score: 0.511894344237182

Round 4:
Best params: {'model__alpha': 1e-05}
Best CV score: 0.5144725899146992

Round 5:
Best params: {'model__alpha': 1e-05}
Best CV score: 0.49487426310329663

```
[ ]: # Hyperparameter tuning for RBF SVM (SVC) NON-LINEAR
from sklearn.svm import SVC

svc_rbf_scores = []

for i in range(5):
    print(f"\nRound {i+1}:")

    # Build pipeline
    svc_pipeline = Pipeline([
        ("prep", full_preprocess),
        ("model", SVC(
            kernel="rbf",
            probability=False,
        ))
    ])

    # Hyperparameter grid
    param_grid_svc = {
        "model__C": [10, 100],
        "model__gamma": [1e-4, 0.001],
    }

    # GridSearchCV with GroupKFold
    svc_grid = GridSearchCV(
        estimator=svc_pipeline,
        param_grid=param_grid_svc,
        cv=gkf,
        scoring="f1_macro",
        n_jobs=-1
```

```

)

svc_grid.fit(X_train_val, y_train_val, groups=groups)

svc_rbf_scores.append(svc_grid.best_score_)

print("Best params:", svc_grid.best_params_)
print("Best CV score:", svc_grid.best_score_)

```

```

[179]: # Hyperparameter tuning for KNN
from sklearn.neighbors import KNeighborsClassifier

knn_scores = []

for i in range(5):
    print(f"\nRound {i+1}:")

    knn_pipeline = Pipeline([
        ("prep", full_preprocess),
        ("model", KNeighborsClassifier())
    ])

    param_grid_knn = {
        "model__n_neighbors": [7, 9, 11],
        "model__weights": ["uniform", "distance"],
        "model__p": [1],
    }

    # GridSearchCV with GroupKFold
    knn_grid = GridSearchCV(
        estimator=knn_pipeline,
        param_grid=param_grid_knn,
        cv=gkf,
        scoring="f1_macro",
        n_jobs=-1
    )

    knn_grid.fit(X_train_val, y_train_val, groups=groups)

    knn_scores.append(knn_grid.best_score_)

    print("Best params:", knn_grid.best_params_)
    print("Best CV score:", knn_grid.best_score_)

```

Round 1:

Best params: {'model__n_neighbors': 7, 'model__p': 1, 'model__weights': 'uniform'}

Best CV score: 0.5392615154351335

Round 2:

Best params: {'model__n_neighbors': 7, 'model__p': 1, 'model__weights': 'uniform'}

Best CV score: 0.5392615154351335

Round 3:

Best params: {'model__n_neighbors': 7, 'model__p': 1, 'model__weights': 'uniform'}

Best CV score: 0.5392615154351335

Round 4:

Best params: {'model__n_neighbors': 7, 'model__p': 1, 'model__weights': 'uniform'}

Best CV score: 0.5392615154351335

Round 5:

Best params: {'model__n_neighbors': 7, 'model__p': 1, 'model__weights': 'uniform'}

Best CV score: 0.5392615154351335

```
[ ]: from sklearn.preprocessing import LabelEncoder
import numpy as np

le_y = LabelEncoder()
y_train_val_enc = le_y.fit_transform(y_train_val)

categorical_cols = ["Month", "Occupation", "Payment_Behaviour"]
ordinal_cols = ["Credit_Mix", "Payment_of_Min_Amount"]

Credit_Mix_order = ["Bad", "Standard", "Good"]
Payment_of_Min_Amount_order = ["Yes", "No"]

from sklearn.preprocessing import OrdinalEncoder
from sklearn.compose import ColumnTransformer

xgb_preprocess = ColumnTransformer(
    transformers=[
        # categorical: encode but preserve NaN
        ("cat", OrdinalEncoder(
            handle_unknown="use_encoded_value",
            unknown_value=np.nan, # unseen categories → NaN
            encoded_missing_value=np.nan # existing NaN → keep NaN
        ), categorical_cols),

        # ordinal: encode with your real order, NaN preserved
    ]
)
```



```

("ord", OrdinalEncoder(
    categories=[Credit_Mix_order, Payment_of_Min_Amount_order],
    handle_unknown="use_encoded_value",
    unknown_value=np.nan,
    encoded_missing_value=np.nan
), ordinal_cols),

    ("num", "passthrough", numeric_cols),
    ("loan", "passthrough",
     [col for col in X_train_val.columns if col.startswith("Loan_")]),
],
remainder="drop"
)

import xgboost as xgb
from sklearn.pipeline import Pipeline

pipeline_xgboost = Pipeline([
    ("prep", xgb_preprocess),
    ("model", xgb.XGBClassifier(
        objective="multi:softprob",
        eval_metric="mlogloss",
        n_estimators=300,
        random_state=0,
        tree_method="exact",
        random_state=42
    ))
])

param_grid_xgb = {
    "model__max_depth": [7, 9, 11],
    "model__learning_rate": [0.01, 0.05],
    "model__subsample": [0.8, 1.0],
    "model__colsample_bytree": [0.8, 1.0],
}

xgb_grid = GridSearchCV(
    estimator=pipeline_xgboost,
    param_grid=param_grid_xgb,
    cv=gkf,
    scoring="f1_macro",
    n_jobs=-1
)
xgb_grid.fit(X_train_val, y_train_val_enc, groups=groups)

print("Best params:", xgb_grid.best_params_)
print("Best CV score:", xgb_grid.best_score_)

```

```

Exception ignored in: <function ResourceTracker.__del__ at 0x1031b9bc0>
Traceback (most recent call last):
  File "/opt/anaconda3/lib/python3.13/multiprocessing/resource_tracker.py", line
82, in __del__
    File "/opt/anaconda3/lib/python3.13/multiprocessing/resource_tracker.py", line
91, in _stop
    File "/opt/anaconda3/lib/python3.13/multiprocessing/resource_tracker.py", line
116, in _stop_locked
ChildProcessError: [Errno 10] No child processes
/opt/anaconda3/lib/python3.13/site-
packages/joblib/externals/loky/process_executor.py:752: UserWarning: A worker
stopped while some jobs were given to the executor. This can be caused by a too
short worker timeout or by a memory leak.
    warnings.warn(
Exception ignored in: <function ResourceTracker.__del__ at 0x1051e5bc0>
Traceback (most recent call last):
  File "/opt/anaconda3/lib/python3.13/multiprocessing/resource_tracker.py", line
82, in __del__
    File "/opt/anaconda3/lib/python3.13/multiprocessing/resource_tracker.py", line
91, in _stop
    File "/opt/anaconda3/lib/python3.13/multiprocessing/resource_tracker.py", line
116, in _stop_locked
ChildProcessError: [Errno 10] No child processes

Best params: {'model__colsample_bytree': 0.8, 'model__learning_rate': 0.01,
'model__max_depth': 7, 'model__subsample': 0.8}
Best CV score: 0.681010757083283

Exception ignored in: <function ResourceTracker.__del__ at 0x107791bc0>
Traceback (most recent call last):
  File "/opt/anaconda3/lib/python3.13/multiprocessing/resource_tracker.py", line
82, in __del__
    File "/opt/anaconda3/lib/python3.13/multiprocessing/resource_tracker.py", line
91, in _stop
    File "/opt/anaconda3/lib/python3.13/multiprocessing/resource_tracker.py", line
116, in _stop_locked
ChildProcessError: [Errno 10] No child processes
Exception ignored in: <function ResourceTracker.__del__ at 0x10273dbc0>
Traceback (most recent call last):
  File "/opt/anaconda3/lib/python3.13/multiprocessing/resource_tracker.py", line
82, in __del__
    File "/opt/anaconda3/lib/python3.13/multiprocessing/resource_tracker.py", line
91, in _stop
    File "/opt/anaconda3/lib/python3.13/multiprocessing/resource_tracker.py", line
116, in _stop_locked
ChildProcessError: [Errno 10] No child processes
Exception ignored in: <function ResourceTracker.__del__ at 0x104841bc0>
Traceback (most recent call last):
  File "/opt/anaconda3/lib/python3.13/multiprocessing/resource_tracker.py", line

```

```

82, in __del__
  File "/opt/anaconda3/lib/python3.13/multiprocessing/resource_tracker.py", line
91, in _stop
  File "/opt/anaconda3/lib/python3.13/multiprocessing/resource_tracker.py", line
116, in _stop_locked
ChildProcessError: [Errno 10] No child processes
Exception ignored in: <function ResourceTracker.__del__ at 0x1028c5bc0>
Traceback (most recent call last):
  File "/opt/anaconda3/lib/python3.13/multiprocessing/resource_tracker.py", line
82, in __del__
  File "/opt/anaconda3/lib/python3.13/multiprocessing/resource_tracker.py", line
91, in _stop
  File "/opt/anaconda3/lib/python3.13/multiprocessing/resource_tracker.py", line
116, in _stop_locked
ChildProcessError: [Errno 10] No child processes
Exception ignored in: <function ResourceTracker.__del__ at 0x1050d1bc0>
Traceback (most recent call last):
  File "/opt/anaconda3/lib/python3.13/multiprocessing/resource_tracker.py", line
82, in __del__
  File "/opt/anaconda3/lib/python3.13/multiprocessing/resource_tracker.py", line
91, in _stop
  File "/opt/anaconda3/lib/python3.13/multiprocessing/resource_tracker.py", line
116, in _stop_locked
ChildProcessError: [Errno 10] No child processes
Exception ignored in: <function ResourceTracker.__del__ at 0x106dfdbc0>
Traceback (most recent call last):
  File "/opt/anaconda3/lib/python3.13/multiprocessing/resource_tracker.py", line
82, in __del__
  File "/opt/anaconda3/lib/python3.13/multiprocessing/resource_tracker.py", line
91, in _stop
  File "/opt/anaconda3/lib/python3.13/multiprocessing/resource_tracker.py", line
116, in _stop_locked
ChildProcessError: [Errno 10] No child processes
Exception ignored in: <function ResourceTracker.__del__ at 0x102319bc0>
Traceback (most recent call last):
  File "/opt/anaconda3/lib/python3.13/multiprocessing/resource_tracker.py", line
82, in __del__
  File "/opt/anaconda3/lib/python3.13/multiprocessing/resource_tracker.py", line
91, in _stop
  File "/opt/anaconda3/lib/python3.13/multiprocessing/resource_tracker.py", line
116, in _stop_locked
ChildProcessError: [Errno 10] No child processes
Exception ignored in: <function ResourceTracker.__del__ at 0x106ec1bc0>
Traceback (most recent call last):
  File "/opt/anaconda3/lib/python3.13/multiprocessing/resource_tracker.py", line
82, in __del__
  File "/opt/anaconda3/lib/python3.13/multiprocessing/resource_tracker.py", line
91, in _stop

```

File "/opt/anaconda3/lib/python3.13/multiprocessing/resource_tracker.py", line 116, in _stop_locked
ChildProcessError: [Errno 10] No child processes

```
[184]: # Hyperparameter tuning for Random Forest
from sklearn.ensemble import RandomForestClassifier

rf_scores = []

for i in range(5):
    print(f"\nRound {i+1}:")

    # Build pipeline
    rf_pipeline = Pipeline([
        ("prep", full_preprocess),
        ("model", RandomForestClassifier(
            random_state=42
        ))
    ])

    # Hyperparameter grid
    param_grid_rf = {
        "model__n_estimators": [200, 300],
        "model__max_depth": [5, 10, 15],
        "model__min_samples_split": [2],
        "model__min_samples_leaf": [1],
        "model__max_features": ["sqrt"],
    }

    # GridSearchCV with GroupKFold
    rf_grid = GridSearchCV(
        estimator=rf_pipeline,
        param_grid=param_grid_rf,
        cv=gkf,
        scoring="f1_macro",
        n_jobs=-1
    )

    rf_grid.fit(X_train_val, y_train_val, groups=groups)

    rf_scores.append(rf_grid.best_score_)

    print("Best params:", rf_grid.best_params_)
    print("Best CV score:", rf_grid.best_score_)
```

Round 1:

Best params: {'model__max_depth': 15, 'model__max_features': 'sqrt',
'model__min_samples_leaf': 1, 'model__min_samples_split': 2,
'model__n_estimators': 200}
Best CV score: 0.6778683014279057

Round 2:
Best params: {'model__max_depth': 15, 'model__max_features': 'sqrt',
'model__min_samples_leaf': 1, 'model__min_samples_split': 2,
'model__n_estimators': 200}
Best CV score: 0.6778683014279057

Round 3:
Best params: {'model__max_depth': 15, 'model__max_features': 'sqrt',
'model__min_samples_leaf': 1, 'model__min_samples_split': 2,
'model__n_estimators': 200}
Best CV score: 0.6778683014279057

Round 4:
Best params: {'model__max_depth': 15, 'model__max_features': 'sqrt',
'model__min_samples_leaf': 1, 'model__min_samples_split': 2,
'model__n_estimators': 200}
Best CV score: 0.6778683014279057

Round 5:
Best params: {'model__max_depth': 15, 'model__max_features': 'sqrt',
'model__min_samples_leaf': 1, 'model__min_samples_split': 2,
'model__n_estimators': 200}
Best CV score: 0.6778683014279057

Exception ignored in: <function ResourceTracker.__del__ at 0x104eb9bc0>
Traceback (most recent call last):
 File "/opt/anaconda3/lib/python3.13/multiprocessing/resource_tracker.py", line
82, in __del__
 File "/opt/anaconda3/lib/python3.13/multiprocessing/resource_tracker.py", line
91, in _stop
 File "/opt/anaconda3/lib/python3.13/multiprocessing/resource_tracker.py", line
116, in _stop_locked
ChildProcessError: [Errno 10] No child processes
Exception ignored in: <function ResourceTracker.__del__ at 0x107039bc0>
Traceback (most recent call last):
 File "/opt/anaconda3/lib/python3.13/multiprocessing/resource_tracker.py", line
82, in __del__
 File "/opt/anaconda3/lib/python3.13/multiprocessing/resource_tracker.py", line
91, in _stop
 File "/opt/anaconda3/lib/python3.13/multiprocessing/resource_tracker.py", line
116, in _stop_locked
ChildProcessError: [Errno 10] No child processes
Exception ignored in: <function ResourceTracker.__del__ at 0x1060d1bc0>

```

Traceback (most recent call last):
  File "/opt/anaconda3/lib/python3.13/multiprocessing/resource_tracker.py", line
82, in __del__
    File "/opt/anaconda3/lib/python3.13/multiprocessing/resource_tracker.py", line
91, in _stop
    File "/opt/anaconda3/lib/python3.13/multiprocessing/resource_tracker.py", line
116, in _stop_locked
ChildProcessError: [Errno 10] No child processes
Exception ignored in: <function ResourceTracker.__del__ at 0x102b45bc0>
Traceback (most recent call last):
  File "/opt/anaconda3/lib/python3.13/multiprocessing/resource_tracker.py", line
82, in __del__
    File "/opt/anaconda3/lib/python3.13/multiprocessing/resource_tracker.py", line
91, in _stop
    File "/opt/anaconda3/lib/python3.13/multiprocessing/resource_tracker.py", line
116, in _stop_locked
ChildProcessError: [Errno 10] No child processes
Exception ignored in: <function ResourceTracker.__del__ at 0x106a3dbc0>
Traceback (most recent call last):
  File "/opt/anaconda3/lib/python3.13/multiprocessing/resource_tracker.py", line
82, in __del__
    File "/opt/anaconda3/lib/python3.13/multiprocessing/resource_tracker.py", line
91, in _stop
    File "/opt/anaconda3/lib/python3.13/multiprocessing/resource_tracker.py", line
116, in _stop_locked
ChildProcessError: [Errno 10] No child processes
Exception ignored in: <function ResourceTracker.__del__ at 0x104fd1bc0>
Traceback (most recent call last):
  File "/opt/anaconda3/lib/python3.13/multiprocessing/resource_tracker.py", line
82, in __del__
    File "/opt/anaconda3/lib/python3.13/multiprocessing/resource_tracker.py", line
91, in _stop
    File "/opt/anaconda3/lib/python3.13/multiprocessing/resource_tracker.py", line
116, in _stop_locked
ChildProcessError: [Errno 10] No child processes
Exception ignored in: <function ResourceTracker.__del__ at 0x106dc1bc0>
Traceback (most recent call last):
  File "/opt/anaconda3/lib/python3.13/multiprocessing/resource_tracker.py", line
82, in __del__
    File "/opt/anaconda3/lib/python3.13/multiprocessing/resource_tracker.py", line
91, in _stop
    File "/opt/anaconda3/lib/python3.13/multiprocessing/resource_tracker.py", line
116, in _stop_locked
ChildProcessError: [Errno 10] No child processes
Exception ignored in: <function ResourceTracker.__del__ at 0x107429bc0>
Traceback (most recent call last):
  File "/opt/anaconda3/lib/python3.13/multiprocessing/resource_tracker.py", line
82, in __del__

```

File "/opt/anaconda3/lib/python3.13/multiprocessing/resource_tracker.py", line 91, in _stop

File "/opt/anaconda3/lib/python3.13/multiprocessing/resource_tracker.py", line 116, in _stop_locked

ChildProcessError: [Errno 10] No child processes

[]:

[220]: *# Select Xgboost as the final model and evaluate on the test set*

```
best_xgb = xgb_grid.best_estimator_
```

```
# Fit on ALL train_val data
```

```
best_xgb.fit(X_train_val, y_train_val_enc)
```

```
# Predict test set (after fitting your best XGB model)
```

```
y_pred = best_xgb.predict(X_test)
```

```
y_pred_proba = best_xgb.predict_proba(X_test)
```

```
label_map = {0: "Good", 1: "Poor", 2: "Standard"}
```

```
y_pred_str = np.array([label_map[x] for x in y_pred])
```

[]: *# Confusion Matrix*

```
# Not normalized confusion matrix
```

```
labels = ["Good", "Poor", "Standard"]
```

```
cm = confusion_matrix(y_test, y_pred_str, labels=labels)
```

```
disp = ConfusionMatrixDisplay(confusion_matrix=cm, display_labels=labels)
```

```
disp.plot(cmap="Blues", values_format="d")
```

```
plt.title("Confusion Matrix XGBoost")
```

```
plt.show()
```

```
# Normalize confusion matrix
```

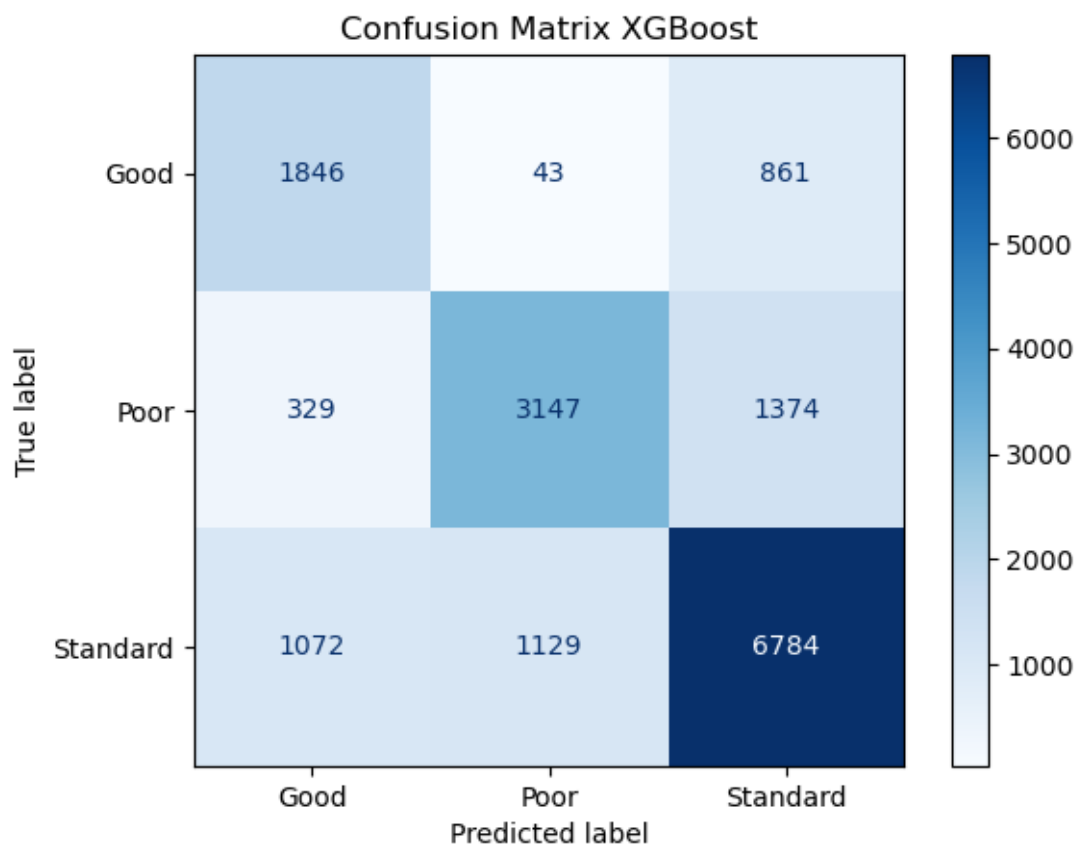
```
cm = confusion_matrix(y_test, y_pred_str, labels=labels, normalize="true")
```

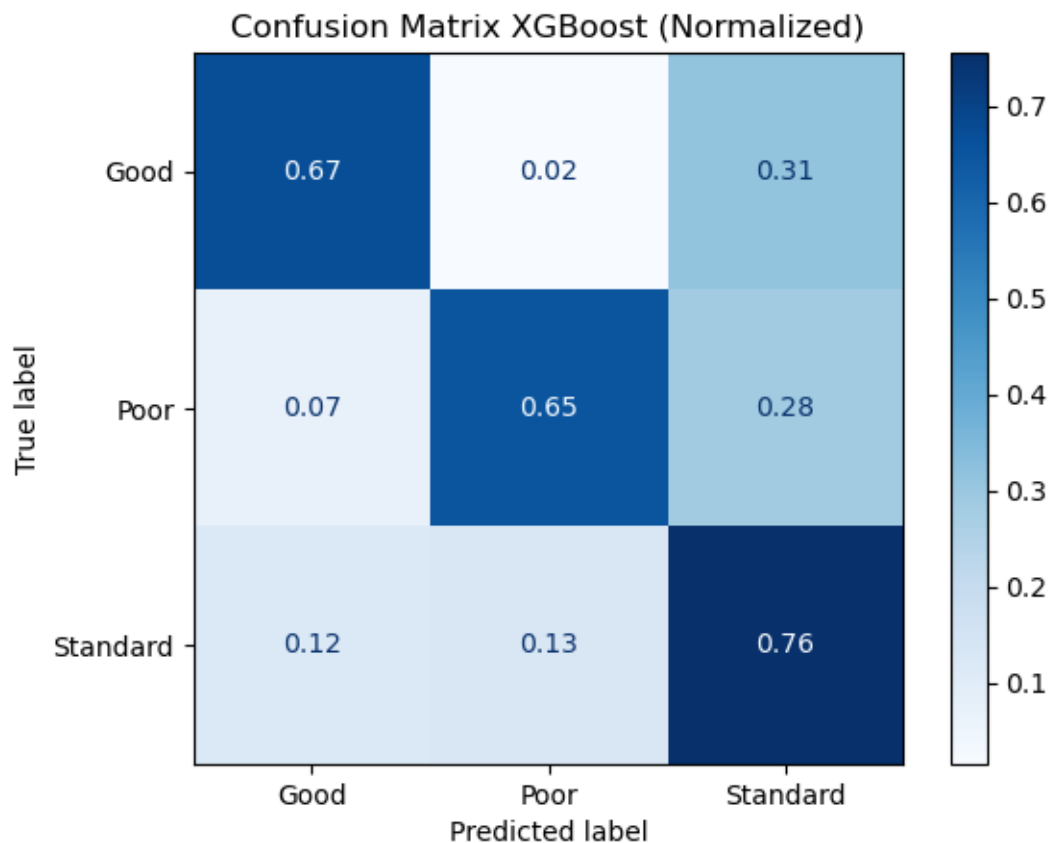
```
disp = ConfusionMatrixDisplay(confusion_matrix=cm, display_labels=labels)
```

```
disp.plot(cmap="Blues", values_format=".2f")
```

```
plt.title("Confusion Matrix XGBoost (Normalized)")
```

```
plt.show()
```





```
[ ]: from sklearn.metrics import classification_report, accuracy_score,
      ↪ balanced_accuracy_score

print("Accuracy:", accuracy_score(y_test, y_pred_str))
print("Balanced Accuracy:", balanced_accuracy_score(y_test, y_pred_str))

print("\nClassification Report:")
print(classification_report(y_test, y_pred_str))
```

Accuracy: 0.7100994874886946

Balanced Accuracy: 0.6917249593503142

Classification Report:

	precision	recall	f1-score	support
Good	0.57	0.67	0.62	2750
Poor	0.73	0.65	0.69	4850
Standard	0.75	0.76	0.75	8985
accuracy			0.71	16585

macro avg	0.68	0.69	0.69	16585
weighted avg	0.71	0.71	0.71	16585

```
[ ]: # ROC Curves for multi-class

from sklearn.preprocessing import label_binarize
from sklearn.metrics import roc_curve, auc
import matplotlib.pyplot as plt
import numpy as np

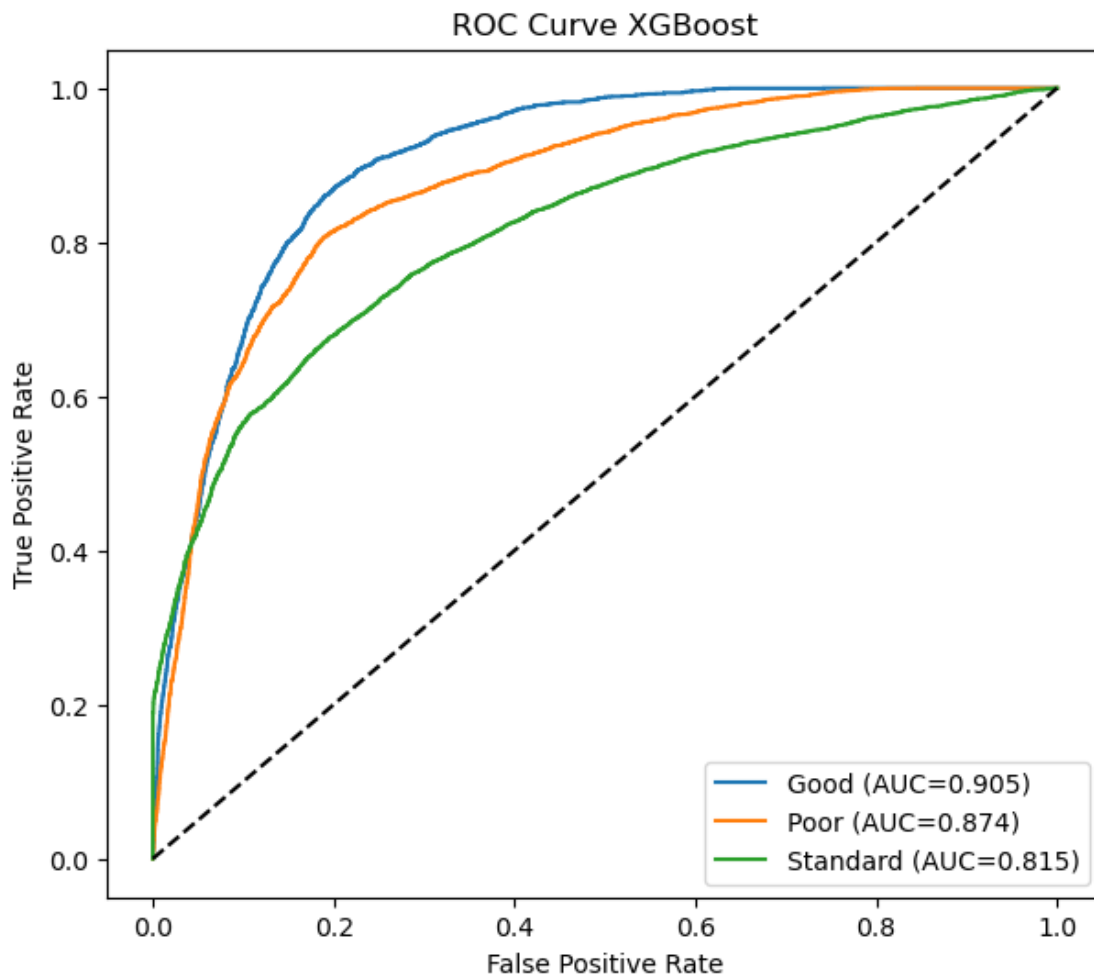
labels = ["Good", "Poor", "Standard"]

y_test_bin = label_binarize(y_test, classes=labels) # shape (n,3)

plt.figure(figsize=(7,6))

for i, cls in enumerate(labels):
    fpr, tpr, _ = roc_curve(y_test_bin[:, i], y_pred_proba[:, i])
    roc_auc = auc(fpr, tpr)
    plt.plot(fpr, tpr, label=f"{cls} (AUC={roc_auc:.3f})")

plt.plot([0,1],[0,1],"k--")
plt.xlabel("False Positive Rate")
plt.ylabel("True Positive Rate")
plt.title("ROC Curve XGBoost")
plt.legend()
plt.show()
```



```
[226]: # Global feature importance from XGBoost
xgb_model = best_xgb.named_steps["model"]
feature_names = best_xgb.named_steps["prep"].get_feature_names_out()

import pandas as pd

importance_types = ["weight", "gain", "cover", "total_gain", "total_cover"]
dfs = []

xgb_model = best_xgb.named_steps["model"]
feature_names = best_xgb.named_steps["prep"].get_feature_names_out()

for imp in importance_types:
    scores = xgb_model.get_booster().get_score(importance_type=imp)

    # dataframe
```

```

df = pd.DataFrame(scores.items(), columns=["feature", imp])

# xgboost feature name f12
df["feature"] = df["feature"].apply(lambda x: feature_names[int(x[1:])])

# top 15
df = df.sort_values(imp, ascending=False).head(15)

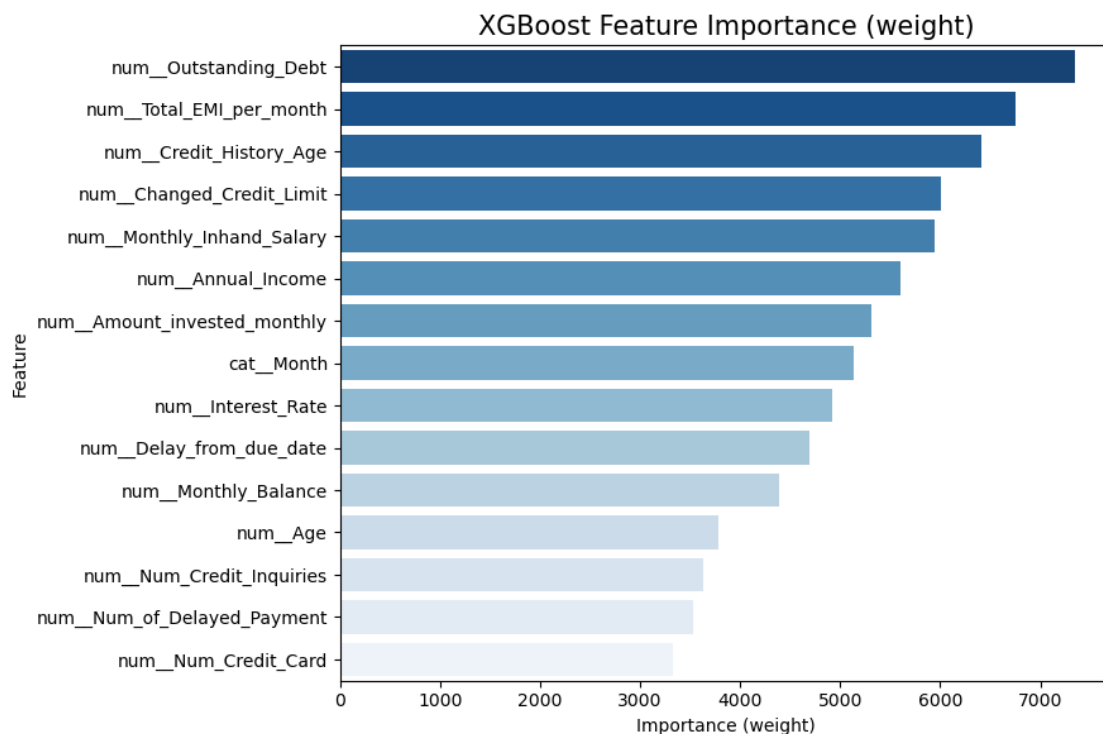
#
plt.figure(figsize=(9, 6))
sns.barplot(data=df, x=imp, y="feature", palette="Blues_r")
plt.title(f"XGBoost Feature Importance ({imp})", fontsize=15)
plt.xlabel(f"Importance ({imp})")
plt.ylabel("Feature")
plt.tight_layout()
plt.show()

```

/var/folders/hs/fgtpkc7n7gg59wjyjcgmqv80000gn/T/ipykernel_33457/3683721086.py:27: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `y` variable to `hue` and set `legend=False` for the same effect.

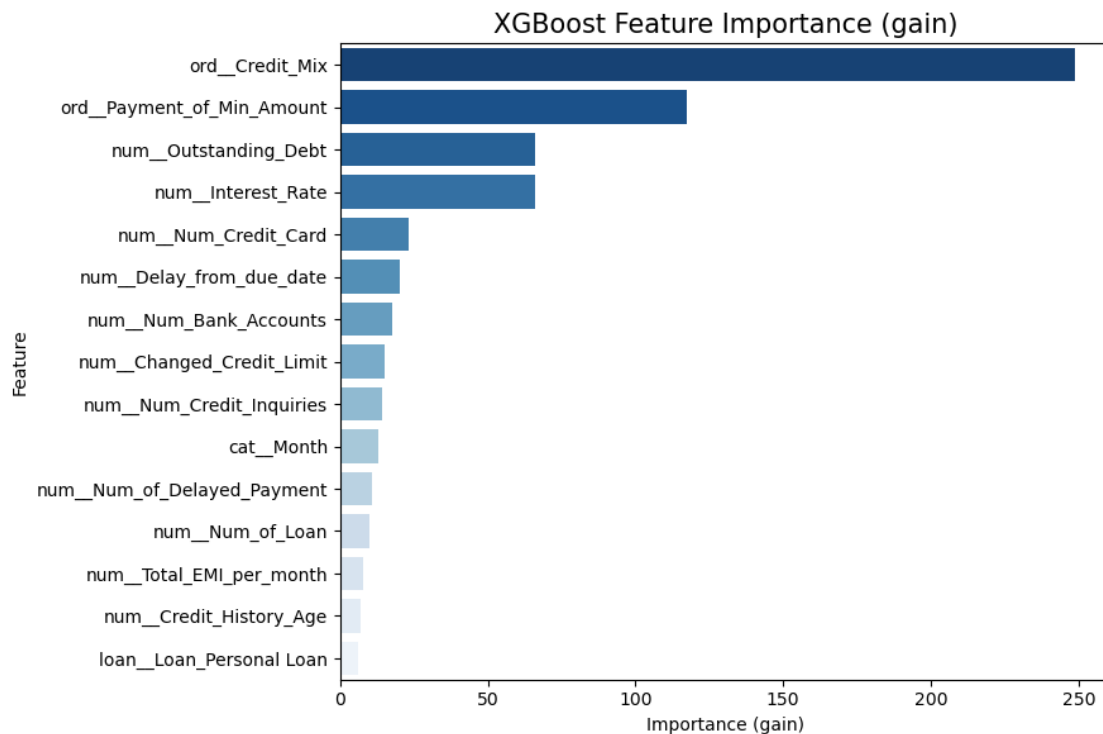
```
sns.barplot(data=df, x=imp, y="feature", palette="Blues_r")
```



```
/var/folders/hs/fgtpkc7n7gg59wjyjcgmqv80000gn/T/ipykernel_33457/3683721086.py:2
7: FutureWarning:
```

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `y` variable to `hue` and set `legend=False` for the same effect.

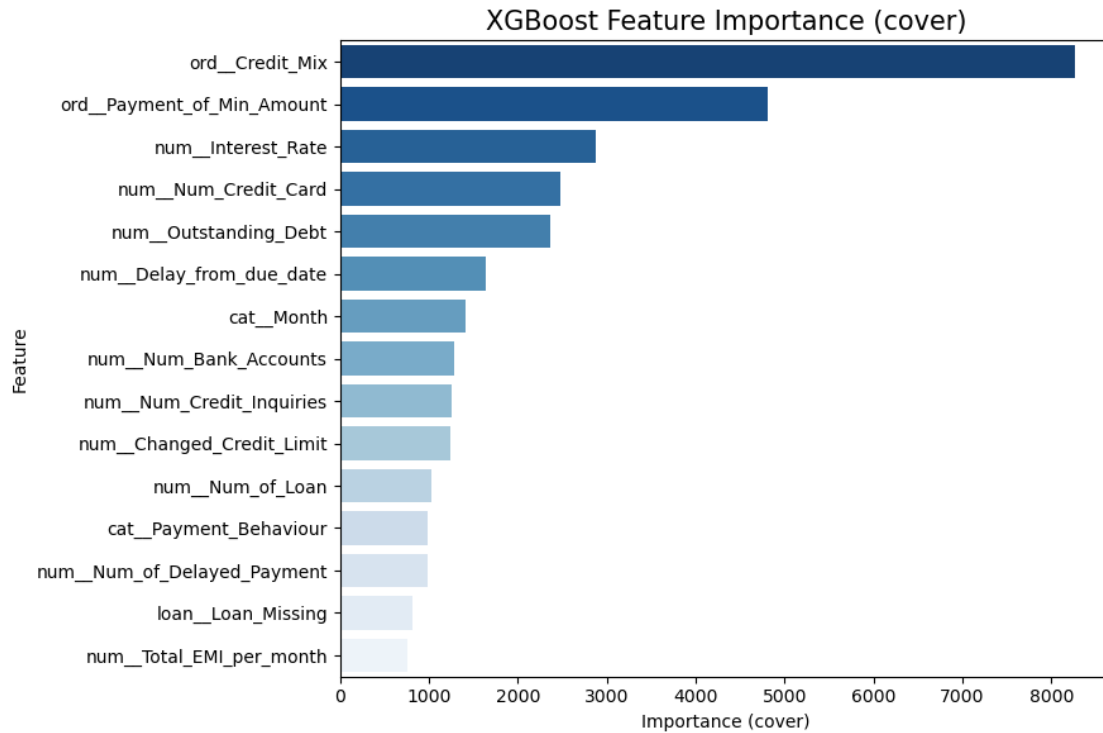
```
sns.barplot(data=df, x=imp, y="feature", palette="Blues_r")
```



```
/var/folders/hs/fgtpkc7n7gg59wjyjcgmqv80000gn/T/ipykernel_33457/3683721086.py:2
7: FutureWarning:
```

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `y` variable to `hue` and set `legend=False` for the same effect.

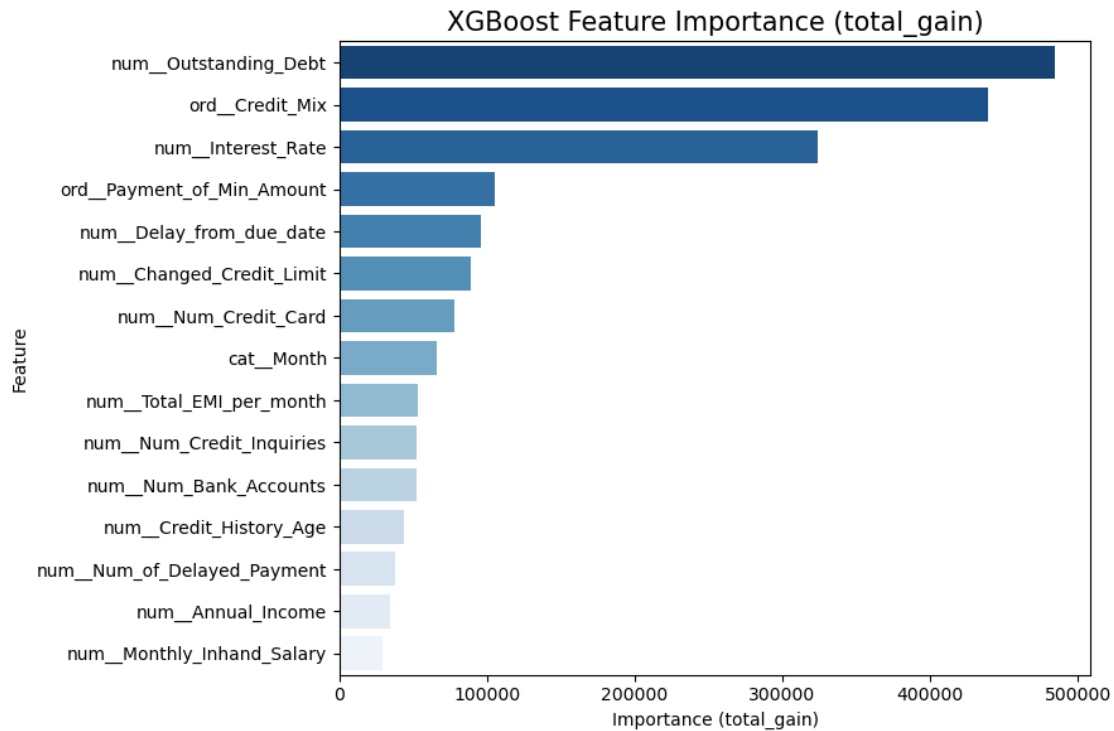
```
sns.barplot(data=df, x=imp, y="feature", palette="Blues_r")
```



/var/folders/hs/fgtpkc7n7gg59wjygcjmqv80000gn/T/ipykernel_33457/3683721086.py:2
7: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `y` variable to `hue` and set `legend=False` for the same effect.

```
sns.barplot(data=df, x=imp, y="feature", palette="Blues_r")
```



```
/var/folders/hs/fgtpkc7n7gg59wjygcjmqv80000gn/T/ipykernel_33457/3683721086.py:2
7: FutureWarning:
```

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `y` variable to `hue` and set `legend=False` for the same effect.

```
sns.barplot(data=df, x=imp, y="feature", palette="Blues_r")
```

